EEE4113F Final Report

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Prepared for:
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August 20, 2026

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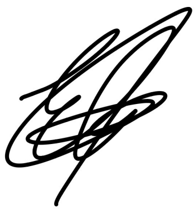

August 20, 2026

Ebrahim Bhyat

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Boitseko Senamela

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August 20, 2026

Joshua Smith

Date

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Chapter 1

Introduction

1.1 Background

The Antarctic Marginal Ice Zone (MIZ) is the transitional region between open ocean and consolidated sea ice, characterised by complex interactions between wave action, ice-floe collisions, and wind-driven drift. Understanding the dynamics of this region is critical to improving models of sea ice extent, ocean-atmosphere heat exchange, and the broader impacts of climate change on polar ecosystems. However, traditional observation methods are severely limited by the logistics of manned expeditions and the spatial resolution constraints of satellite remote sensing, leaving a critical gap in high-fidelity, in-situ measurements of sea ice behaviour at the floe scale.

This problem was conceived during UCT D-School ideation sessions in which the client, Ms. Robyn Verrinder of the UCT Department of Electrical Engineering, presented the problem of continuous Antarctic MIZ monitoring, and the project team synthesised an autonomous buoy platform as the proposed engineering solution

Autonomous surface buoys offer a practical solution to this data gap, enabling continuous multi-modal measurement without the cost and risk of sustained human presence in the MIZ. Existing platforms such as the UCT SHARC V3.0 and the Estonian LainePoiss buoy have demonstrated the viability of this approach, but face limitations in sample rate, onboard processing pipelines that discard high-frequency data, and communications architectures that are economically unviable at scale [6, 7].

The SIREN (Sea Ice Remote Environmental Node) system is developed in response to these limitations. It is designed as a low-power, deployable platform capable of continuous high-frequency ice-floe dynamics measurement, raw data storage, and low-cost wireless data retrieval, built to survive the extreme mechanical and thermal conditions of Antarctic all within a budget of R1500.

1.2 Objectives

The primary objective of the SIREN project is to design, build, and validate an autonomous ice buoy system capable of continuous monitoring of ice-floe dynamics in the Antarctic MIZ. The specific objectives of the project are:

- To develop a sensing subsystem capable of sampling six-degree-of-freedom IMU data at high frequency and logging raw measurements to non-volatile storage for land-based post-processing.

- To develop a power management subsystem capable of sustaining operation under Antarctic sub-zero conditions through duty cycling and thermal management.
- To develop a watertight, thermally insulated enclosure capable of withstanding wave action, ice-floe collisions, and temperature differentials down to $-40\,^{\circ}\mathrm{C}$ while maintaining static and dynamic stability.
- To develop a low-cost communications and networking subsystem for infrastructure-free data retrieval using Bluetooth Low Energy and ESP-NOW mesh networking.
- To validate each subsystem independently through defined Acceptance Test Procedures prior to integration.

1.3 System Requirements

The top-level system requirements for SIREN are derived from the user requirements provided by Ms. Robyn Verrinder, UCT Department of Electrical Engineering. The system shall:

- Continuously measure ice-floe dynamics including linear acceleration and angular velocity at a minimum sample rate sufficient to resolve wave frequencies up to 0.5 Hz.
- Record all sensor data in raw, unprocessed format for retrieval and land-based post-processing.
- Determine and transmit the geographic position of the buoy at regular intervals.
- Operate continuously for a minimum deployment duration under Antarctic environmental conditions including temperatures down to $-40\,^{\circ}\mathrm{C}$.
- Withstand the mechanical loading of wave action, ice-floe collisions, and wind gusts without structural failure or fluid ingress.
- Provide a low-cost data retrieval mechanism that does not rely on satellite infrastructure for routine operation.
- Remain statically and dynamically stable during aquatic operation to maintain consistent sensor orientation.

1.4 Scope & Limitations

The scope of this project encompasses the design, implementation, and acceptance testing of four subsystems: sensing, power management, enclosure, and communications. Each subsystem is designed and validated independently by an individual team member, with inter-subsystem interfaces defined to enable integration.

The following limitations apply to the current project phase:

- Full system integration testing is beyond the scope of this report. Each subsystem is validated independently through its defined ATPs.

- Antarctic field deployment and environmental validation are not conducted. All testing is performed in a laboratory and outdoor urban environment in Cape Town.
- Communications range is validated in a controlled environment and not under Antarctic sea ice field conditions.

1.5 Report Outline

This report is structured as follows. Chapter 3 presents the sensing subsystem, covering IMU, GNSS, and SD card design choices, firmware implementation, and ATP results. Chapter 4 presents the power management subsystem, covering thermal management strategy, battery selection, and duty cycle architecture. Chapter 5 presents the enclosure subsystem, covering structural geometry, material selection, waterproofing strategy, and stability analysis. Chapter 6 presents the communications and networking subsystem, covering the BLE data retrieval architecture and ESP-NOW mesh network implementation. Chapter 7 concludes the report with a summary of results across all subsystems and recommendations for the deployment hardware revision.

The complete design files, firmware, and schematics for the SIREN project are openly available on our GitHub repository.

Chapter 2

Literature Review

2.1 Sensing

2.1.1 The Antarctic MIZ as a measuring environment

Sea ice dynamics in the Antarctic Marginal Ice Zone (MIZ) are governed by a complex interplay of wind stress, ocean currents, wave forcing and internal ice stresses [8]. Critically, the net stress acting on a floe cannot be derived from these mean forcing values alone. High-frequency fluctuations including wind gustiness, turbulent ocean eddies, wave motion and floe-floe collisions contribute significantly to the overall stress regime [8]. This establishes that a sensing system deployed on an Antarctic ice floe must be capable of resolving both slowly varying environmental conditions and the rapid, high-frequency dynamic behaviour of the floe itself.

Field measurements by Alberello et al. demonstrate that Antarctic pancake ice drift reaches instantaneous speeds of up to 0.75 m/s during polar cyclones, with floe velocity and ice deformation successfully tracked using high-accuracy GPS position fixes at 15-minute intervals [9]. This establishes a minimum positional sampling requirement and confirms GPS as a necessary sensing component. However, the buoys carried no onboard meteorological or wave sensors, with wind forcing instead inferred from ERA5 satellite reanalysis data rather than measured locally [9]. This limitation motivates the inclusion of direct environmental sensing in SIREN, providing localised measurements that eliminate the need for external data sources. Furthermore, Herbers et al. also found that using an onboard GPS is a viable low-cost solution to measuring the horizontal wave orbital displacements [10]. However, the vertical sea surface displacements were not well resolved by the GPS position receivers which implies that a more robust sensing solution must be developed.

To capture the high-frequency wave-in-ice motion that was absent from prior buoy deployments, an Inertial Measurement Unit (IMU) is identified as a cost-effective and capable solution. Rabault et al. demonstrate that IMUs can reliably measure wave properties in landfast ice, providing all three components of linear acceleration and angular velocity — characterising the full 6-DOF motion of a floating ice floe [11]. Importantly, a recommendation from project stakeholders notes that complex onboard signal processing such as Fast Fourier Transforms is constrained by the limited instruction sets of embedded processors. Following this guidance, SIREN transmits raw IMU data for post-processing on land, consistent with the approach validated by Herbers et al. who demonstrated that fusing raw inertial data with GPS velocity measurements yields well-resolved wave elevation estimates [10].

Beyond motion sensing, Roach et al. identify the connections between sea ice growth, melt, and upper-ocean heat and stratification as critically important yet poorly observed quantities in the Antarctic [8]. The absence of direct temperature measurement in prior autonomous deployments further compounds this gap. This absence highlights the need to incorporate dedicated air temperature, water temperature and humidity sensors directly onto the node, enabling localised environmental characterisation that satellite reanalysis products cannot provide at the spatial and temporal scales relevant to individual floe dynamics.

2.1.2 Existing Sensing Platforms and Their Limitations

UCT SHARC V3.0

The work by Noyce in [6] is directly focused on developing a buoy which was used to measure waves-in-ice in the Antarctic MIZ which incorporated the use of IMUs and an STM32 microcontroller. The primary method for measuring waves-in-ice relies on a low-cost 6-DoF IMU, specifically the ICM20649. The IMUs monitors the physical oscillation of the floe. The MCU used ultimately used a polling system to communicate with the IMU after an I2C DMA program on the STM32 was deemed unreliable. The 6-DoF raw acceleration and angular velocity data was nominally sampled at 100Hz. The polling sampling system had the disadvantage of drawing resources from the MCU CPU during the duration of the IMU sampling period. To monitor the drift of the ice floes, the platform uses a continuous GPS/GNNS receiver, particularly the UBlox Neo 9M GNSS/GPS receiver which measured air temperature and atmospheric pressure. Running the GPS continuously avoids the delays and potential failures associated with cold or warm start lock-ons in extreme temperatures. The GPS data was streamed into a circular buffer which ensured that the system was able to obtain the latest geographical data when it was requested. The collected data is processed on-board and the summarised data is then sent to be studied. The summary statistics were sent using an Iridium satellite modem where the MCU communicated with the modem by using a UART interface using simple text-based AT commands. This has the disadvantage of the data sent being susceptible to software bugs or poor signal processing which can make the statistics unreliable. Furthermore, sending summary statistics instead of raw-data can risk the loss of potentially insightful high-frequency data. This limitation suggests the validity of transmitting raw data rather than processed data.

LainePoiss

The LainePoiss (LP) buoy, developed by Alari et al., offers a complementary comparison point to the SHARC platform. The LP employs a 9-DoF Xsens MTi-3 AHRS sensor with an internal extended Kalman filter for full Earth-referenced sensor fusion, sampling at 50 Hz [7]. While this enables directional wave parameter extraction, the onboard processing pipeline introduces significant constraints on transmitted data. Over satellite networks, the LP is limited to processed summary statistics, sacrificing the raw high-frequency content that is critical for resolving phenomena such as floe-floe collisions [7]. Furthermore, double integration of IMU data amplifies low-frequency gyroscope noise, requiring a custom denoising procedure to compensate for lost spectral energy — a correction that is only possible in post-processing [7]. The SHARC buoy faces analogous constraints, relying on aggressive decimation and fixed-length FFTs due to MCU memory limitations, and similarly restricts satellite transmission to 100-byte summary packets [6]. Collectively, these platforms demonstrate that onboard processing inevitably involves irrecoverable data loss or correction uncertainty.

2.1.3 Sensor Choices

To capture the high-frequency wave motion that drives sea ice fracture and modifies floe size distribution, a sensor capable of measuring three-dimensional acceleration and angular velocity is required [8]. Rabault et al. (2016) demonstrated that Inertial Measurement Units (IMUs) offer a cost-effective method to reliably extract these full six-degree-of-freedom wave kinematics in ice-covered waters [11]. Furthermore, recent autonomous platforms, such as the SHARC V3.0 and the LainePoiss buoy, have successfully utilised low-cost microelectromechanical system (MEMS) IMUs to extract valid wave spectra in polar and sub-polar environments [6, 7]. However, a critical limitation of commercial MEMS IMUs in this application is their susceptibility to random gyroscope noise, which introduces severe low-frequency drift when the raw acceleration data is double-integrated to calculate wave displacement [7]. Given the documented risk of irrecoverable information loss associated with onboard processing, future iterations can mitigate this vulnerability by offloading uncompressed IMU data for land-based post-processing, where complex spectral analysis can be executed without embedded computational constraints.

To calculate the strain rates and deformation of the ice field, the autonomous node requires highly precise horizontal positioning to track floe velocities [9]. Alberello et al. (2020) demonstrated that standard GPS receivers are highly effective for this application, successfully tracking instantaneous pancake ice drift speeds of up to 0.75 m/s during polar cyclones [9]. Furthermore, Hunt (2020) validates that while three-dimensional satellite positioning degrades at polar latitudes, the impact on horizontal accuracy remains trivial, cementing its utility for two-dimensional drift tracking [12]. However, because GPS satellite orbits are inclined at 55 degrees relative to the equator, Dilution of Precision (DOP) severely degrades at Antarctic latitudes, resulting in delayed solution convergence and poor vertical resolution [12]. To mitigate this geometric limitation and ensure rapid lock-on in freezing conditions, the design replaces legacy GPS-only modules with a multi-constellation GNSS receiver (the UBlox NEO-M9N) operating in continuous mode, which capitalizes on the overlapping availability of GPS, Galileo, and GLONASS satellites to expedite accurate positioning [5, 12].

While surface-level environmental conditions can be captured using low-power digital sensors (such as the BME280), extracting temperature profiles using analog components introduces significant electrical challenges in polar environments. Zuo et al. (2018) demonstrated that operating analog-to-digital converters (ADCs) and amplification circuits at temperatures approaching $-50°$C introduces measurable voltage drift [13]. Mitigating these extreme-cold analog errors requires the implementation of complex, dynamically-calculated temperature correction algorithms on the microcontroller [13]. Consequently, utilizing fully integrated, factory-calibrated digital environmental sensors is highly preferable for the surface node to ensure measurement stability without incurring additional computational overhead.

2.1.4 Data Handling

Autonomous polar nodes typically transmit only processed summary statistics via Iridium SBD due to the strict 340-byte payload limit and high financial cost per message [4]. However, as demonstrated by both the SHARC V3.0 and LainePoiss platforms, onboard processing inevitably involves irrecoverable data loss, whether through aggressive decimation, fixed-length FFTs, or spectral averaging that destroys transient time-domain events such as floe-floe collisions [6, 7]. Furthermore, Herbers et al. demonstrated that fusing raw inertial data with GPS velocity measurements on land yields well-resolved wave elevation estimates that onboard processing cannot replicate [10]. Therefore, a solution is necessary that departs from the summary statistics approach, directly transmitting raw 6-DoF IMU data and GPS fixes, accepting the higher transmission cost in exchange for a complete, uncompromised dataset available for post-processing on land.

2.2 Power Supply in Antarctic Conditions

Reliable power is essential for Antarctic data collection. Historically, expeditions relied on risky diesel generators and manual measurements. Today, automated instruments have eliminated these human and environmental risks, but they introduce new challenges in safely and reliably powering remote, battery-dependent systems [14].

Keeping a battery-powered module able to survive extreme conditions is an important aspect in the longevity of the deployments of any autonomous module destined for the Antarctic. When placed in extremely cold conditions, both battery capacity and performance decrease dramatically [15]. To achieve the goals of sustained operations, it is vital that the design of the power module can mitigate the effects of the extreme cold. To achieve this, a highly integrated approach spanning cell selection, hybrid energy architectures and algorithmic thermal management is needed, since conventional lithium-ion batteries naturally suffer from capacity degradation, increased internal resistance and sluggish chemical kinetics [14, 15].

2.2.1 Battery Cell Format and Chemistry

The physical and chemical makeup of a battery can dictate its baseline resistance to freezing conditions. Abdelrahman et al. [15] and Rampal et al. [16] confirm this occurs due to the cold negatively affecting battery physics and results in reduced ionic conductivity and lithium plating within the cell.

When comparing rechargeable Li-Ion cells, Lithium Manganese Oxide ($LiMn_2O_4$) chemistries generally exhibit lower internal resistance and better sub-zero performance than Lithium Cobalt Oxide ($LiCoO_2$) chemistries, due to the kinetic advantage derived from three-dimensional spinel structure present in $LiMn_2O_4$ cells, which possess lower charge transfer resistance[15]. However, physical formats of cells can overshadow chemical properties, such as when $LiCoO_2$ outperformed $LiMn_2O_4$ in 18650 and 21700 formats[15].

For non-rechargeable cells, we can look towards Jacobson's 2021 SHARC Buoy. Jacobson [17] initially utilised a Lithium Thionyl Chloride ($LiSOCl_2$) battery, which has a theoretical resistance to the cold and high specific energy, but suffered from unexpected increases in internal resistance that resulted in brownouts and resets. Jacobson then replaced them with Lithium Iron Disulphide ($LiFeS_2$) batteries for better low temperature stability at the cost of lower specific energy [17]. By choosing a reliable and robust type of battery, the additional systems needed to sustain its performance can be minimised, which aids in the goal of designing any low-cost battery-powered module destined for the Antarctic.

2.2.2 Hybrid Super-capacitor Architectures

Standard lithium-ion batteries struggle not only with sustained discharge in sub-zero environments but also with initiating a cold start [18]. To mitigate the initial voltage drop and power constraints, hybrid energy storage systems that pair a supercapacitor with a standard lithium-ion cell can be employed.

In this architecture, the supercapacitor supplies high-current power to trigger phase-change preheating for an initial period (e.g., the first 30 seconds). A dual-MOSFET switching strategy then seamlessly transitions the load to the primary battery to continue module operation without voltage drops [18]. While this approach successfully offloads the high-stress preheating phase from the battery, contrasting with methods that rely entirely on the battery's own depleted power for self-heating [1], supercapacitors suffer from limited nominal capacitance and degradation over time [18]. Consequently, while effective for brief cold-start assistance, a hybrid approach alone is insufficient for long-term, remote deployments and requires supplementary thermal management.

2.2.3 Active Heating Structures

Beyond cold-start assistance, maintaining the battery within a favourable temperature window is critical to mitigating the effects of the cold. The literature suggests two distinct formats to achieve this: external and internal heat generation.

External heating strategies often utilise self-powered structures, such as Polyimide (PI) heating films placed between cells within an insulated enclosure [1]. While these films provide ample heat to ease a cold start and maintain operating temperatures, they introduce significant drawbacks for low-cost, low-power modules. These limitations include increased design complexity, added weight, parasitic energy consumption, and the risk of uneven thermal propagation across the cell pack.

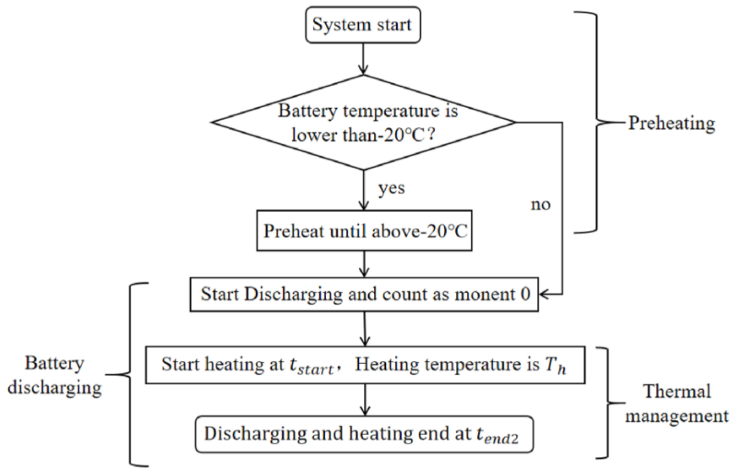

Figure 2.1: Flowchart of external heating system implementation [1].

Conversely, internal heating relies on generating heat from within the cell itself. Integrating a nickel foil heating element allows a battery's internal temperature to rapidly increase from -20°C to 0°C in 20 seconds [19]. However, this approach necessitates a complete battery redesign and introduces unclear safety implications [20]. To address this without altering the physical cell, Bidirectional Pulse Current

(BPC) methods can be utilised, which rely on the battery's own internal resistance to generate Joule heat. This system is highly efficient and mitigates risks such as overvoltage, significantly improving battery longevity - especially in the remote regions where automated modules are deployed [20].

2.2.4 Algorithmic Estimation and Control

Maximising the efficiency of any self-heating hardware requires a robust Battery Management System (BMS) capable of real-time estimation and control. Accurately determining the State of Charge (SOC) and State of Power (SOP) in sub-zero environments is challenging but critical. Advanced filtering techniques, such as combining an Extended Kalman Filter (EKF) with Adaptive Recursive Least Squares (ARLS), have proven highly accurate for real-time state measurement [1].

Once battery states are accurately estimated, the BMS must optimise the heating strategy to ensure the energy spent on heating does not outweigh the usable capacity gained. Approaches to this optimisation vary; integer genetic algorithms have successfully increased discharge durations by over 90% by strictly governing heat timing [1], while neural networks paired with dichotomy optimisation algorithms excel at calculating exact pulse amplitudes for internal Joule heating without violating voltage limits [20]. For a low-cost, low-power Antarctic module, implementing these complex electro-thermal models presents a computational bottleneck. To mitigate this, replacing heavy real-time processing with simplified offline calculations and look-up tables offers a viable compromise between thermal efficiency and processing overhead.

2.3 Communication Network and Signal Propagation

Establishing a reliable communication link in marine Antarctic data collection is complicated by the region's absence of conventional infrastructure and low available bandwidth [21]. This section details the use of a multi-hop communication setup as well as an adaptive approach to the selection of available radio access technologies.

2.3.1 Radio Access Technologies in Polar Regions

Existing communication strategies in both Arctic and Antarctic regions rely mainly on satellite technologies such as the Iridium satellite system[22], the issue however with the transmission of data using these satellite technologies is that they come with high financial costs associated with them. The SHARC is a buoy developed with the Iridium 9603 modem as a transceiver, sending data in packets of 340 bytes at a cost of R2 [6]. The non-scalability of its operation however becomes apparent once we consider that in order to get the most information from the buoy, sampling would need to happen continuously and data transmission frequently. During a power acceptance test the SHARC buoy spent 2.4 days constantly working, alternating between 5 minutes of sensing and burst of satellite transmission, until it reached 700-transmissions. This would mean that for some hypothetical buoy operating almost continuously between sensing and transmission, the transmission costs for a single buoy would cost close to R20,000 for a single month and approach R100, 000 over the span of a couple months. Koo et al. investigate the implementation of internet of extreme things technologies in providing a framework for adaptive sensor networks in polar conditions, as well as software defined radio technologies enabling dynamic frequency allocation. Koo et al. investigate the implementation of long-range wireless communications for Antarctic climate monitoring which encompass an IoET device transmitting data collected by sensors to a base station through the gateway [21]. The authors note the IoET device and gateway support both wired and wireless access technologies such as Ethernet, Wi-Fi, LoRa, and ZigBee [21], and as such, the need for a method of access selection arises.

2.3.2 Multi-criteria Decision Making models in access selection

In heterogenous networks, signal quality is frequently uncertain due to interference and environmental changes [23]. As such, in order to maintain stable links, communication systems must move beyond simple signal-strength metrics and adopt a hybrid decision model such as the technique for order of preference by similarity to ideal solution (TOPSIS) in order to facilitate vertical handover between access technologies[24]. Guo et al. argue that in the Antarctic, traditional multi-criteria decision making (MCDM) models that rely on precise values are often insufficient because network attributes such as available bandwidth or delay are frequently inaccurate due to interference, signal fluctuation, and high mobility [23]. Taking this into consideration, a fuzzy logic variant of TOPSIS can be considered. Yu et al. explore an intuitionistic Fuzzy TOPSIS framework which captures ambiguous environmental data when fuzzy network attributes such as bandwidth or jitter cannot be obtained [25]. The authors also explore a utility-based fuzzy TOPSIS framework utilised as a way to perform energy-efficient network selection where the fuzziness of battery life and energy consumption is integrated into the TOPSIS ranking process [25].

2.3.3 Physical Link Propagation

Elgharbi et al. note that the physical structure of the communication link must account for unique propagation singularities which are likely to degrade the quality of the radio signal [2]. According to the authors, these singularities are primarily driven by two factors: the low height of the antennas as well as the dynamic nature of an ice floe as the buoy's support structure [2]. The authors suggest that an antenna height of 3 meters is necessary for the gateway node to the central server (on the order of transmission across 10km) given that the height of the antenna is a significant factor in attenuation. They find that when the antenna is positioned too close to the water, the transmitted signal undergoes significant attenuation due to diffraction and the distortion of the fresnel ellipsoid. While an antenna height of 10m would decrease attenuation by over 20dB, it is challenging to place antennas at such a height on a reasonably sized buoy [2].

Additionally, while conventional wisdom assumes Antarctic ice acts as a high-absorption material for HF radio wave[26], with an intensity loss of approximately 85% at a 45 degree incident angle, Liu et al. provide evidence that using rocky terrain as a reflector bypasses this absorption thus enabling two-hop F-region multi-path reflections for group ranges exceeding 1800 km.

2.3.4 Multi-hop Mesh Topographies

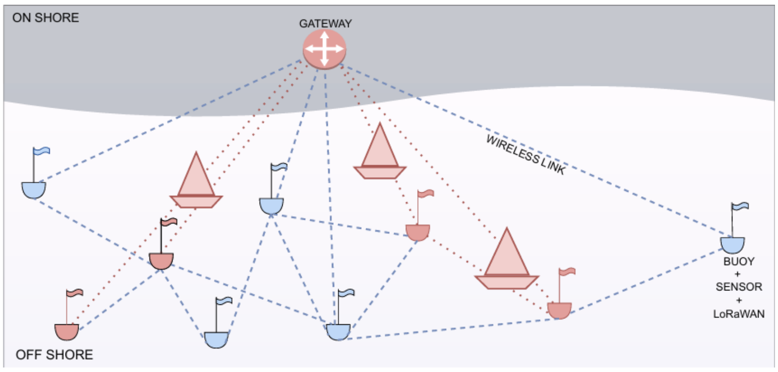

Figure 2.2: Multi-hop relay architecture in a maritime LoRaWAN mesh network [2].

In scenarios involving ice floes, which mimic maritime environments, the literature advocates for a LoRaWAN mesh network [2] . A fully decentralized mesh introduces a series of technical implications as the gateway node to the central server not only has to function as a relay, but in the work by Koo et al. it is responsible for data storage and power control along with data collection [21]. Because direct node-to-gateway communication often fails at extended ranges due to interference and obstacles, Elgharbi et al. advocate for a multi-hop approach utilizing a single relay node. The authors note that this approach has been shown to triple throughput and reduce the packet loss ratio (PLR) from 78% to near 0% up to distances of 5km.

2.4 System Enclosure and Dynamics

An increasing need for data in Marginal Ice Zone's (MIZ) has arised due to 'in situ' observations and data regarding the sea ice floe dynamics created by waves in the region [5, 4]. Older implementation of these forms of research has been limited due to lack of continuous data as researchers have to physically conduct experiments and acquire data. However, modern satellite imaging has several drawbacks despite its wide coverage of the region as it is not able to collect data on a finer scale. Human based approaches in manually collecting data during expeditions to Antarctic provide the greatest accuracy but is expensive due to logistics and is limiting as only a small part of the region can be observed. The only viable solution for this is deployment of autonomous buoys situated on the ice surface to do so.

However, the sea ice surface is dynamic and subject to rapid change, hence creating a gap in information due to lack of observation in the surrounding ocean. A system capable of bridging this gap with the ability to survive this complex, dynamic environment will therefore present its own mechanical challenges. To address these limitations, the system should be able to survive a range of temperatures due to the difference between the ocean and ice which drops to about $-40°$C [27, 3, 28], as well as mitigate mechanical forces created by waves, ice collisions and strong gusts. Due to the dynamic nature of the region as well as rapid changes of state due to ice melting or breaking apart, the buoy should not only survive both the ice and ocean but the transition between the two states as well [3, 5]. One therefore has to consider the geometry of the structure alongside the weight distribution to allow for it to remain statically upright once in the water to keep data consistent and accurate.

The internal section of the structure also needs to be considered, as it provides housing for components needed to allow the system to carry out its purpose. These dimensions are heavily influenced by the power supply and requirements, communication hardware (ie. antenna), insulation and overall cost. The enclosure should not be big in size to allow for convenient deployment whilst having sufficient space for the power supply as well as allowing the structure to remain buoyant. There has been a shift towards open-source enclosures due to their low cost [4], as these often smaller in size and lightweight solutions are able to keep logistics costs at a minimum whilst allowing for a less invasive approach.

2.4.1 Enclosure Geometry

The geometry of the enclosure is dictated by its dynamic nature on the ice and in the water as well as the systems deployment and purpose. Several existing projects on the ice surface are deployed vertically through the ice thus needing the enclosure to adopt a standard cylindrical form such that it can be lodged into the ice. This includes the Ice Tethered Profiler (ITP) which is embedded into the ice through a 25cm hole[29]. Another similar approach to this includes the Polar Ice-Based Ecological Observation Buoy which is cylindrical in shape and is deployed through a hole in the ice, with additional support to prevent the enclosure from being crushed by any colliding ice[27].

A more conventional approach to the problem at hand involves multi-sensor enclosures which remain above the surface of the ice. These structures utilise the increased buoyancy from spherical geometries as well as greater housing space created for internal components. The Ice-Based Buoy consists of a rigid sphere which makes up the main body of the system, and ensures it remains upright on the ice with a flat steel base [3].

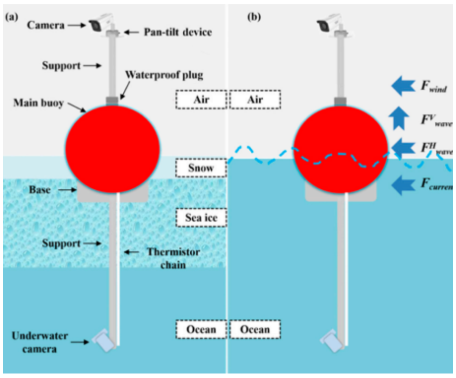

Figure 2.3: Diagram showing the structure of the Ice-Based Buoy [3].

Projects prioritising affordability and portability lean towards simple enclosures of small sizes. The OpenMetBuoy-v2021, for example uses a rectangular box design for its enclosure to allow for easier deployment on the ice surface [4]. Due to the basic design, it lacks stability within the ocean when the internal components are heavy. A more modified approach to this same design, dubbed the 'Floatenstein', utilised the same rectangular box structure with added styrofoam strapped to all sides as well as a metal keel running lengthwise at the bottom of the enclosure [4]. This allowed for it to maintain buoyancy despite the increased weight from the internal components as well as increased vertical stability. This proves that simple modifications can increase the functionality of the enclosure itself.

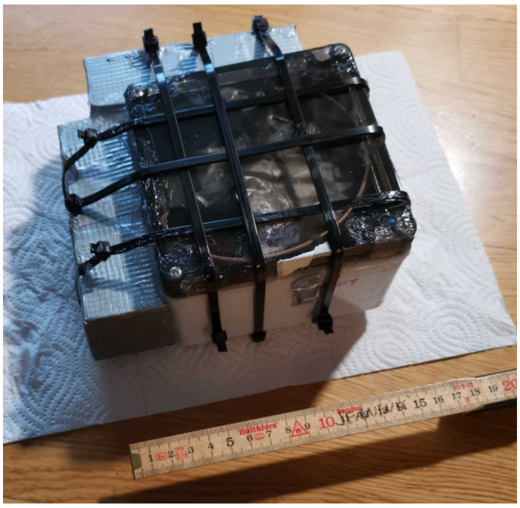

Figure 2.4: 'Floatenstein' Enclosure Design [4].

Lastly, the UCT SHARC Buoy V3.0 utilises a geometry in which the sensor housing is elevated above the ice surface to negate the effects of overwash from waves [5]. This however introduces rotational dynamics to the system which need to be accounted for and must therefore be mathematically modelled. This elevation is advantageous as it protects components from the colder temperatures of the ice surface and bolts the structure into the ice through the rigid stand at its base.

Figure 2.5: 3D Render of the UCT SHARC Buoy 3.0 [5].

2.4.2 Material Selection and Environmental Effects

Prolonged time within the extreme conditions of the MIZ requires for the enclosure to withstand extremely low temperatures as well as external force from the environment. This raises concern with regards to the materials used for the enclosure as well as sealing techniques to allow the structure to maintain its integrity and functionality during deployment.

External Protection

The external shells of most existing applications are made up of polyvinyl chloride (PVC) as this creates a rigid outer layer for the structure [3] as well as surlyn ionomer foam which covers the internal components and aids in shock absorption [29].

Materials for Dynamic Environment Operation

To protect the integrity of internal components, the main body of the structure should be able to withstand various amounts of pressure as well as corrosion from the seawater. To provide this protection against the environmental conditions, various components such as the power supply is enclosed in a stainless steel housing to prevent water from infiltrating into any cavities and affecting the system [5]. Pressure is not as much of a problem due to the buoy floating on the surface of the water and not being exposed to immense hydrostatic pressure in the depths of the ocean. However, it can be noted that titanium alloy is used to line the top and bottom of waterproof casings due to its lightweight and strong nature [3].

Housing Waterproofing and Sealing Techniques

The method which stood out most with regards to sealing was the use of epoxy glue in the 'Floatenstein' enclosure [4]. This allows for the enclosure to be completely watertight, whilst providing a low cost solution to the problem at hand. On a more commercial scale, they are more reliant on 'O' rings, which provide equal pressure on two halves of the enclosure that are joined together and are applied to heavier and durable housings [4].

2.4.3 Hydrodynamics

When transitioning from the ice surface to the ocean due to melting, ice collisions or other environmental factors, it is vital for the buoy itself to remain upright such that data collection can remain consistent and ultimately sinking of the enclosure itself. This is achieved through the combination of weight distribution and enclosure geometry such that the metacenter remains above the enclosures center of gravity [27].

Ballasting is the main approach when considering ways to increase buoy stability in water whilst attenuating oscillations from wave currents. For structures which remain close to or on the water surfaces, light counterweights are tethered to the bottom of the enclosure [27]. In more simple approaches such as the 'Floatenstein', a simple keel was fixated to the bottom of the enclosure to increase its stability without a tethered counterweight below it.

Additionally, when operating in the ice, the enclosure should mimic the response of actual sea ice such that data can remain consistent throughout its deployment. Failure to do so will result in collected data being incorrect. To achieve this, researchers have done extensive hydrodynamic testing of mock disk-shaped platforms using polypropylene due to its similar density to actual sea ice [5]. During testing it was discovered that the platform was able to perfectly follow the motion of waves when the wavelength of the wave was equal to or greater than three times the diameter of the mock platform [5]

2.4.4 Sea Ice Interfacing

Both mechanical and thermal challenges arise when the enclosure interfaces with sea ice, especially along boundaries between the ice and the ocean.

A prominent phenomenon which occurs is solar absorption from the buoy, which in turn heats up the surrounding ice around the enclosure. This melting can reduce stability and cause the structure to topple over [29, 27]. Ways in which this can be prevented is through increases area over which heat is dissipated. This has been done with the Ecological Observation Buoy which was placed upon a wooden base to prevent direct contact with the ice as well as distribute the weight and thermal effect across a larger piece of ice [27]. Similarly, the ITP is also placed on a wooden base to decrease its direct contact with the ice as well as the yellow housing of the enclosure which allows for mitigates solar radiation whilst allows it to be discernible from the surrounding snow [29].

The enclosure should also protect the internal components from external forces exerted on the structure. This is seen in the ITP's surlyn ionomer foam which allows for internal shock absorption [29]. This ties in closely with material interactions, particularly when enclosures need to be retrieved from the environment. This is shown in the UCT SHARC Buoy which allows for easy retrieval due to the high-density polyurethane (HDPE), which does not freeze to the metal stand keeping it upright [5]. This in conjunction with the stand which keeps the sensor housing above the sea ice surface and keeps the structure in place due to spikes which remain embedded in the ice allows for optimal interfacing as well as convenient retrieval once the deployment period has ended.

2.5 Conclusion

This chapter has reviewed the literature underpinning the four subsystems of an autonomous ice buoy designed for deployment in the Antarctic Marginal Ice Zone. Across all subsystems, a consistent theme emerges: existing autonomous platforms are individually capable but collectively flawed, each constrained by a specific combination of sensing gaps, power limitations, communication costs, and mechanical fragility that prevents sustained, high-fidelity data collection in one of the most under-observed regions on Earth.

The sensing literature establishes that meaningful characterisation of ice floe dynamics requires simultaneous capture of high-frequency motion, positional drift, and localised environmental conditions that satellite reanalysis products cannot resolve at the spatial and temporal scales relevant to individual floes [8, 9]. Prior platforms validate the use of low-cost MEMS IMUs for wave-in-ice measurement but are constrained by onboard processing pipelines that result in irrecoverable loss of high-frequency data during satellite transmission [6, 7]. The power literature demonstrates that conventional lithium-ion chemistries degrade significantly at sub-zero temperatures, and that sustained polar operation requires a combination of robust cell chemistry, hybrid supercapacitor architectures, and internal heating strategies governed by simplified offline algorithms [17, 20, 15]. The communications literature highlights that satellite telemetry alone is financially unsustainable at high sampling rates, and that a multi-hop LoRaWAN mesh topology offers a viable near-field link provided antenna height and propagation conditions are carefully managed [2, 21]. Finally, the enclosure literature confirms that survival in the MIZ demands careful attention to geometry, material selection, waterproofing, and hydrodynamic stability, with the transition between ice and ocean presenting a particularly demanding design condition [4, 5, 27].

Together, these findings define a clear set of design requirements for any autonomous sensing node intended for long-duration Antarctic deployment, and motivate the integrated, cross-subsystem approach taken in this project.

Chapter 3

Sensing Subsystem (ALMMOH017)

Prepared by Maarij Alam

3.1 Introduction

This subsystem involves designing a hardware and software solution for the collection and storage of ice-floe dynamics and position data. The dynamics data is saved to be post-processed, and the position data is to be transmitted periodically.

3.2 Requirements and Specifications

3.2.1 User Requirements

The user requirements relevant to this subystem are provided by Ms. Robyn Verrinder from UCT Dept. of Electrical Engineering. The user requirements pertaining to the sensing subsystem are provided in the list below:

- Ice floe dynamics need a way to be tracked
- Ice floe dynamics must be recorded for retrieval
- The data must be raw data that must be post-processed
- There needs to be a way to track the position of the bouy
- The system should be power efficient

3.2.2 Requirements Analysis

Using these user requirements, the following functional requirements (FR), specifications (SP), and Acceptance Test Procedures (ATP) were created.

Table 3.1: Functional Requirements of the Sensing Subsystem

FR ID	Functional Requirement
FR-1	The subsystem shall be controlled by a microcontroller capable of managing multiple concurrent sensor interfaces and peripheral communication protocols.
FR-2	The subsystem shall measure ice floe motion by sampling three-axis linear acceleration and three-axis angular velocity using a 6-DoF MEMS IMU at a minimum rate of 100 Hz.
FR-3	The subsystem shall determine the geographic position of the ice floe by receiving and parsing GNSS position fixes at a minimum rate of 1 Hz.
FR-4	The subsystem shall log all raw IMU and GPS data to a local non-volatile storage medium in a structured, retrievable format.
FR-5	The subsystem shall provide a beacon data packet containing latitude, longitude, battery state of charge, and a UTC timestamp to the network subsystem at regular intervals.
FR-6	The subsystem shall characterise the active power consumption of the sensing subsystem to inform duty cycle planning for deployment.
FR-7	The subsystem shall ensure all sensor interfaces operate without data corruption or bus conflicts when running concurrently.

Table 3.2: System Specifications with associated FRs and ATPs

SP ID	Specification	FR ID	ATP ID
SP-1	The STM32F401 microcontroller shall operate at 3.3 V and a clock frequency of 84 MHz, with SPI, UART, and I2C peripherals simultaneously active.	FR-1	ATP-1
SP-2	The LSM9DS1 IMU shall be sampled at a minimum of 100 Hz, outputting raw 6-DoF data (acceleration in m/s^2, angular velocity in deg/s) with the magnetometer disabled.	FR-2	ATP-2, ATP-3, ATP-4
SP-3	The u-blox NEO-6M GPS receiver shall output valid $GPRMC NMEA sentences over UART at 1 Hz, with position accuracy within 10 m of a known reference coordinate under open sky conditions.	FR-3	ATP-5
SP-4	The SD card shall log a minimum of 100 rows per second in CSV format (`timestamp_ms, ax, ay, az, gx, gy, gz, lat, lon`) with zero missing samples over a 10-minute continuous logging period.	FR-4	ATP-6
SP-5	The beacon data packet (lat, lon, battery_%, timestamp — 13 bytes total) shall be passed to the network subsystem every 15 minutes via the agreed interface.	FR-5	ATP-8
SP-6	The sensing subsystem active current draw shall be measured with all peripherals operational. Steady-state current shall be documented and peak current shall not exceed 150 mA on the 3.3 V sensor rail.	FR-6	ATP-7
SP-7	All three sensor interfaces (SPI for IMU, UART for GPS, SPI for SD card) shall operate concurrently without bus contention, data corruption, or system fault over a minimum 15-minute integration test period.	FR-7	ATP-9

Table 3.3: Acceptance Test Procedures — Sensing Subsystem

ATP ID	FR(s)	Test Procedure	Pass Criteria
ATP-1	FR-1, FR-7	**MCU and Peripheral Initialisation.** *(Integration test.)* Connect the Black Pill to a laptop via USB and upload the integrated firmware. Open the Serial Monitor at 115200 baud and observe the boot messages. Measure the 3.3 V rail with a multimeter.	All three messages appear: `IMU init OK`, `SD init OK`, `GPS init OK`. No fault flags printed. 3.3 V rail: 3.2–3.4 V.

Continued on next page

ATP ID	FR(s)	Test Procedure	Pass Criteria
ATP-2	FR-2	**IMU Static Accuracy.** *(Unit test.)* Place board flat and record 100 `az_raw` samples via the ArduinoIDE serial monitor. Tilt to 45° using a phone inclinometer and repeat. Hold flat and still for 30 seconds after a 40-second re-settle period and record maximum `az_filt`.	Flat mean: $9.81 \pm 0.5\,\mathrm{m/s^2}$. 45° mean: $6.94 \pm 0.5\,\mathrm{m/s^2}$. `az_filt` stationary: $< 0.1\,\mathrm{m/s^2}$.
ATP-3	FR-2	**IMU Tap Detection.** *(Unit test.)* Place board flat and still. Record 10 seconds of baseline data to SD card, then apply three sharp vertical taps. Copy CSV to laptop and run Python analysis script to compute baseline RMS of `az_filt` and identify spike events exceeding the threshold.	Spike in `az_filt` exceeds $2\times$ baseline RMS across ≥ 2 consecutive samples per tap event.
ATP-4	FR-2	**IMU Wave Simulation.** *(Unit test.)* Shake board vertically at 1 Hz using a 60 BPM metronome for 30 seconds. Copy CSV to laptop and run Python Welch PSD script on `az_filt` to estimate the power spectral density and identify the dominant frequency component.	Dominant PSD peak: $1.0 \pm 0.1\,\mathrm{Hz}$. SNR $\geq 10\,\mathrm{dB}$ above noise floor.
ATP-5	FR-3	**GPS Position Accuracy.** *(Unit test.)* Take board outdoors. Wait for fix. Record coordinates and compare to Google Maps reference via Haversine formula. Count fix updates over 60 seconds.	Fix acquired within 5 minutes. Fix rate: 60 ± 3 per minute. Position error $< 10\,\mathrm{m}$.
ATP-6	FR-4	**SD Card Data Integrity.** *(Unit test.)* Run recording firmware for 10 minutes. Copy `siren.csv` to laptop and run Python validation script to check row count, detect null or corrupted fields, and compute the achieved sample rate from row timestamps.	Row count: $60\,000 \pm 300$ rows ($100\,\mathrm{Hz} \times 600\,\mathrm{s}$). Zero null or corrupted fields. Sample rate: $\geq 100\,\mathrm{Hz}$.
ATP-7	FR-6	**Active Current Draw.** *(Unit test.)* Place multimeter in series on 3.3 V sensor rail. Run full integrated firmware for 2 minutes and record steady-state and peak current.	Total system steady-state current measured and documented. Peak current $\leq 150\,\mathrm{mA}$.
ATP-8	FR-5	**Beacon Packet Verification.** *(Unit test.)* Upload firmware with beacon interval reduced to 60 seconds. Acquire GPS fix outdoors. Record three consecutive `BEACON` transmissions using the ArduinoIDE serial monitor.	Three beacons at $60 \pm 2\,\mathrm{s}$ intervals. Packet: 13 bytes, sync `0xBE`, valid coordinates.

Continued on next page

ATP ID	FR(s)	Test Procedure	Pass Criteria
ATP-9	FR-1, FR-7	**System Integration Endurance.** *(Integration test.)* Run all three sensors simultaneously for 15 minutes. Monitor Serial Monitor for errors. Validate CSV with Python script.	Zero resets or crashes. Zero null or corrupted values. Sample rate: $\geq 70\,\text{Hz}$. GPS fix coverage: $> 80\%$.

3.3 Design Choices

The sensing subsystem requires a microcontroller to configure and interface with the IMU and GPS receiver, manage data flow to the SD card, and prepare beacon packets for handoff to the network subsystem. The microcontroller must support SPI, UART, and I2C communication protocols concurrently, include a floating point unit (FPU) for efficient sensor data handling, and provide sufficient flash memory and SRAM for firmware and circular buffer implementation. Due to budget and timeline constraints, only off-the-shelf development boards were considered.

3.3.1 Microcontroller

Three candidate platforms were evaluated against the following figures of merit: hardware FPU availability, DMA support for high-frequency sensor streaming, peripheral concurrency, ecosystem maturity for polar buoy applications, and unit cost.

Table 3.4: Microcontroller Candidate Comparison

Component	Flash	SRAM	FPU	DMA	SPI/UART/I2C	Price
Raspberry Pi Pico	2MB	264KB	No	Yes	Yes	R82.90
STM32F401 Black Pill	256KB	64KB	Yes	Yes	Yes	R110.00
ESP32 DevKit	4MB	520KB	No	Limited	Yes	R115.90

The **Raspberry Pi Pico** (RP2040) was considered due to its low cost and large SRAM. However, it lacks a hardware FPU, introducing significant software overhead when computing floating point filter operations at 100 Hz.

The **ESP32** was considered for its large flash and wireless capability. However, its Wi-Fi and Bluetooth peripherals consume significant power unsuitable for a battery-constrained polar deployment.

The **STM32F401CE** (Black Pill development board) was selected for its hardware FPU, which is required for floating-point filter operations at 100 Hz without software overhead, and its three independent SPI peripherals, allowing the IMU and SD card to operate on separate buses without contention.

3.3.2 Inertial Measurement Unit

Ice floe motion in the Antarctic MIZ is characterised by a superposition of wave-induced oscillations and slower wind-driven drift, requiring a sensor capable of resolving both high-frequency dynamic motion and low-frequency attitude changes [8]. An Inertial Measurement Unit (IMU) providing three-axis linear acceleration and three-axis angular velocity was identified as the appropriate sensing solution, consistent with the approach validated by Rabault et al. (2016) and adopted by both the SHARC V3.0 and LainePoiss platforms [11, 6, 7].

Table 3.5: IMU Candidate Comparison

Component	DoF	Interface	Max ODR	Temp Range	Price
MPU-6050	6	I2C only	1kHz	-40 to 85°C	R74.47
LSM9DS1	9	SPI/I2C	952Hz	-40 to 85°C	R148.00
ICM-42688	6	SPI/I2C	32kHz	-40 to 85°C	R228.47

The MPU-6050 was eliminated due to its I2C-only interface. As documented in the SHARC V3.0 development, I2C DMA on STM32 microcontrollers is unreliable during high-frequency continuous sampling due to clock stretching conflicts and STM32 I2C silicon errata [6]. SPI was therefore a hard requirement for the IMU interface.

The ICM-42688-P offers superior noise performance and a higher output data rate, however its cost exceeds the per-component budget allocation and breakout boards were not locally available within the project timeline.

The LSM9DS1 was selected as it supports SPI communication, provides 6-DoF motion data at output data rates up to 952 Hz (well above the required 100 Hz), and operates across the full Antarctic temperature range of -40°C to 85°C. The integrated magnetometer is disabled in firmware, as magnetometers are unreliable near the Antarctic magnetic poles due to extreme magnetic interference [6]. Raw 6-DoF acceleration and angular velocity data is sampled at 100 Hz and written directly to the SD card without onboard processing, preserving the complete high-frequency dataset for land-based post-processing consistent with the approach validated by Herbers et al. [10].

3.3.3 GNSS Receiver

Accurate Lagrangian drift tracking requires periodic position fixes to determine floe velocity and displacement over time. Alberello et al. demonstrated that GPS position fixes at 15-minute intervals are sufficient to characterise Antarctic pancake ice drift during polar cyclone events [9]. A GNSS receiver operating at 1 Hz in continuous mode was therefore specified to provide significantly higher temporal resolution than this minimum requirement.

Table 3.6: GNSS Receiver Candidate Comparison

Component	Constellations	Interface	Cold Start	Price
u-blox NEO-6M	GPS only	UART	27s	R154.00
u-blox NEO-M8N	GPS, GLONASS	UART	26s	R557.09
u-blox NEO-M9N	GPS, GLONASS, Galileo, BeiDou	UART	24s	R1,439.00

The NEO-M9N represents the optimal choice for Antarctic deployment, as its multi-constellation support mitigates Dilution of Precision (DOP) degradation at high latitudes caused by the 55-degree orbital inclination of GPS satellites [12]. However, its cost exceeds the available budget allocation.

The u-blox NEO-6M was selected due to budget constraints, at R95 providing sufficient capability for prototype demonstration in Cape Town where polar DOP degradation is not a factor. The receiver operates in continuous mode to avoid cold-start lock-on failures in freezing temperatures, consistent with the approach adopted by the SHARC V3.0 [6]. NMEA sentences are received over UART using DMA with idle-line interrupt detection, minimising CPU load during continuous reception. The single-constellation limitation is acknowledged as a known constraint and a multi-constellation receiver is recommended for any future field deployment.

3.3.4 Data Storage

All raw sensor data is logged to a local microSD card. No onboard signal processing is performed prior to storage. This approach is directly motivated by the finding that prior platforms including the SHARC V3.0 and LainePoiss buoy suffer irrecoverable high-frequency data loss through onboard processing pipelines [6, 7]. Raw data storage ensures that transient events such as floe-floe collisions — which spectral averaging erases — are preserved for land-based analysis.
Data is written in CSV format with the following column structure:

```
timestamp_ms, ax, ay, az, gx, gy, gz, lat, lon
```

The IMU drives the write rate at 100 rows per second. GPS fields are updated at 1 Hz and repeated across 100 rows until the next fix arrives. At this rate, an 8 GB SD card provides approximately 2.3 days of continuous logging. Duty cycling the IMU to a 5-minute active window per hour extends this to approximately 27 days — sufficient for the target deployment duration.

3.4 System Design

The final sensing subsystem consists of an STM32F401CE microcontroller interfacing with an LSM9DS1 IMU over SPI, a u-blox NEO-6M GPS receiver over UART, and a microSD card over SPI. The IMU samples 6-DoF motion data at 100 Hz, buffered in RAM and flushed to the SD card in CSV format. GPS NMEA sentences are parsed non-blocking at 1 Hz, with the latest fix written alongside each IMU sample. A 13-byte beacon packet containing position, battery state, and UTC timestamp is passed to the network subsystem every 15 minutes. The complete sensing subsystem schematic, showing all peripheral connections to the STM32F401CE microcontroller, is provided in Appendix A.

3.4.1 Block Diagram

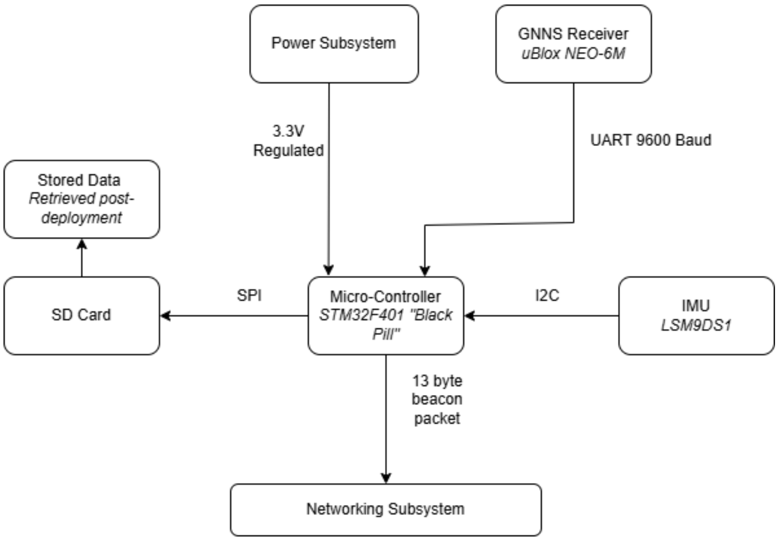

Figure 3.1: Subsystem Block Diagram

3.4.2 Control Flowchart

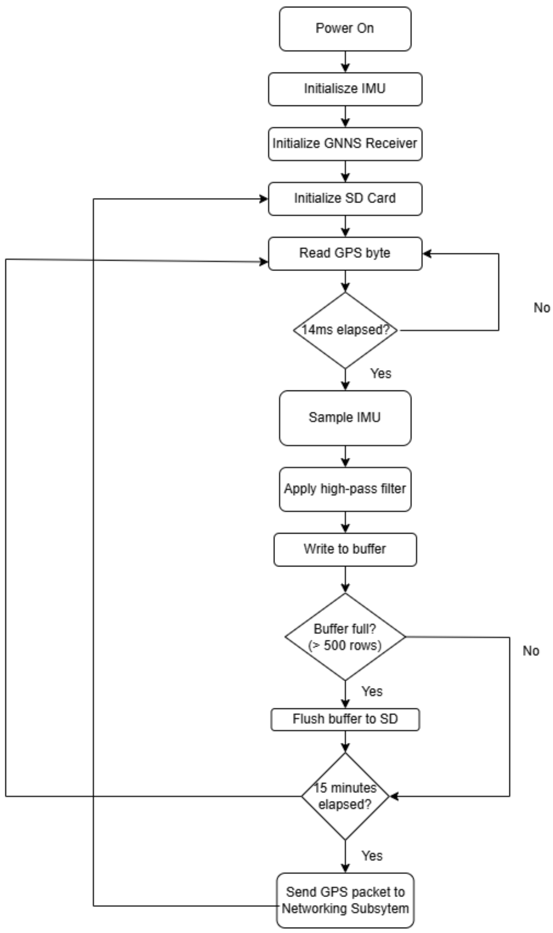

Figure 3.2: System Control Flowchart

3.5 ATP Results

Table 3.7: ATP Results Summary — Sensing Subsystem

ATP ID	Measured Result	Outcome
ATP-1	3.3 V rail: 3.26 V. All boot messages appeared: `IMU init OK`, `SD init OK`, `GPS init OK`. No fault flags.	**PASS**
ATP-2	Flat mean: 9.706 m/s^2. 45° mean: 7.058 m/s^2. Stationary filter max: 0.085 m/s^2.	**PASS**
ATP-3	Peak spike: 12.0 m/s^2. Threshold (2×RMS): $\pm$1.13 m/s^2. All tap events exceeded threshold across $\geq$2 consecutive samples.	**PASS**
ATP-4	Dominant PSD peak: 1.05 Hz. SNR above noise floor: 17.7 dB.	**PASS**
ATP-5	Time to fix: <5 minutes. Fix rate: 59.9 per minute. Position error: 22.3 m (HDOP = 1.6, 6 satellites). *Error attributed to urban multipath. Hardware operating correctly. NEO-M9N recommended for deployment.*	**FAIL**
ATP-6	Row count: 42 600 rows. Null values: 0. Sample rate: 71 Hz via I2C. *SPI migration attempted but unsuccessful due to STM32duino library compatibility issues. I2C retained for prototype. 71 Hz exceeds Nyquist for 0.5 Hz MIZ waves — scientific validity maintained.*	**FAIL**
ATP-7	Steady-state current: 65–75 mA. Peak current: $\sim$100 mA.	**PASS**
ATP-8	Beacon timestamps: t = 62, 122, 182 s. Interval: 60 $\pm$ 2 s. Packet size: 13 bytes. Sync byte `0xBE` confirmed. 900 s deployment interval confirmed by code inspection.	**PASS**
ATP-9	Rows logged: 64 285. Sample rate: 71 Hz. GPS fix coverage: 100%. Zero resets, zero corrupted values. *Timestamp gaps caused by GPS UART blocking — root cause diagnosed and firmware fix documented.*	**PASS**

3.6 Conclusions and Recommendations

The sensing subsystem prototype successfully demonstrated continuous collection and local storage of ice-floe dynamics and position data across seven of nine acceptance tests. The IMU correctly resolved static acceleration, transient tap events, and wave-frequency motion, with a PSD peak of 1.05 Hz and SNR of 17.7 dB confirmed under controlled excitation. The SD card logged continuously for 15 minutes without corruption or bus conflicts, and the beacon packet was correctly formatted and transmitted at the specified 900 s interval.

Two specifications were not met in the prototype implementation. The target 100 Hz IMU sample rate via SPI was not achieved due to compatibility issues between the Adafruit LSM9DS1 driver and the STM32duino SPI HAL implementation. I2C was retained at an achieved rate of 71 Hz, which exceeds the Nyquist frequency for the maximum MIZ wave frequency of 0.5 Hz by a factor of 71, preserving the scientific validity of the prototype. GPS position accuracy of 22.3 m exceeded the 10 m criterion, attributable to urban multipath degradation in Cape Town rather than hardware failure, with an HDOP of 1.6 and six satellites confirming correct receiver operation.

The following recommendations are made for the deployment hardware revision:

SPI IMU interface. SPI communication for the LSM9DS1 should be re-implemented using the STM32 HAL library directly, bypassing the STM32duino abstraction layer. This is expected to resolve the library compatibility issue and achieve the 100 Hz target sample rate, reducing per-sample bus time from approximately 350 µs to 15 µs.

GNSS receiver upgrade. The u-blox NEO-6M should be replaced with the u-blox NEO-M9N for Antarctic deployment. The NEO-M9N supports four concurrent GNSS constellations — GPS, GLONASS, Galileo, and BeiDou — which is critical for reliable operation at Antarctic latitudes. GPS satellites are limited to a 55-degree orbital inclination, meaning that at high southern latitudes the number of visible GPS satellites is significantly reduced and geometric dilution of precision (GDOP) increases [12]. GLONASS, which operates at 64.8-degree inclination, provides substantially better polar coverage, and BeiDou's MEO satellites further improve geometric diversity. The combined multi-constellation fix of the NEO-M9N is therefore not merely a performance improvement but a functional necessity for reliable position tracking in the Antarctic MIZ, where single-constellation GPS fixes may be intermittent or unavailable during periods of low satellite visibility.

Duty cycling firmware. IMU duty cycling should be implemented to extend the continuous logging duration beyond the current 2.3-day limit imposed by the 8 GB SD card at 100 Hz. A 5-minute active window per hour reduces the effective data rate by a factor of 12, extending logging to approximately 27 days — sufficient for the target deployment duration.

GPS blocking fix. The GPS UART blocking issue identified in ATP-9 should be resolved by replacing the blocking `while` loop with a single-byte `if` read, as identified during testing. This fix eliminates timestamp gaps of up to 742 ms and should be validated with a full 30-minute endurance retest.

Chapter 4

Power System (SMTJOS022)

Prepared by Joshua Smith

4.1 Introduction

The power supply is critical to the SIREN system's ability to measure environmental parameters and communicate remotely under Antarctic sub-zero conditions. The initial design centred on Bidirectional Pulse Current (BPC) as the primary thermal management strategy, which uses an alternating square current through the Li-ion battery's internal resistance to implement Joule heating. Power budget analysis revealed this to be incompatible with the deployment time requirement. The architecture was therefore revised to duty-cycled normal operation as the primary strategy, with BPC retained as an emergency cold-recovery mechanism.

4.2 Requirements Analysis

One of the key considerations identified through interviews with Robyn Verrinder is the need for a reliable power delivery system for any autonomous buoy. Most battery chemistries have significant reliability issues in sub-zero conditions that impede operations of any remote module. Another key consideration is the lifespan of any deployment, which has historically been limited to a few weeks. Ideally, Robyn wishes to extend deployments to a couple months to gather more data, especially over winter seasons.

Table 4.1: Requirements of the power subsystem

ID	Requirement
RQ-01	Power System needs to provide stable power for a duration of 1 week.
RQ-02	Power unit must be replaceable and scalable.
RQ-03	System needs to monitor and supply health data out.
RQ-04	System must survive and operate autonomously in sub-zero conditions.
RQ-05	System must safely manage volatile battery chemistries without risk of thermal runaway or lithium plating.
RQ-06	Overall system needs to sufficiently heat Lithium Ion Battery using internal Joule heating.
RQ-07	System needs to implement Bidirectional Pulse Current to prevent net charge/discharge.
RQ-08	System needs to supply constant voltage out.
RQ-09	System needs to communicate downtime with external modules.

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Table 4.1 – continued from previous page

ID	Requirement
RQ-10	System needs to monitor real-time voltage and current to estimate remaining battery capacity.
RQ-11	System must measure ambient temperature to trigger heating cycle.

Table 4.2: Specifications of the power subsystem

ID	Specification	Develops
SP-01	The control system shall operate for a minimum of 7 days (168 hours) on a single 2400 mAh Li-$SOCl_2$ primary cell at a maximum average quiescent current draw of $\leq 14.3\,\text{mA}$ when idle.	RQ-01
SP-02	The Li-$SOCl_2$ and Li-Ion cells shall be housed in removable, accessible battery holders requiring no tools for replacement.	RQ-02
SP-03	The BPC system shall raise the Li-ion battery temperature from $-30\,°\text{C}$ to $0\,°\text{C}$ within 15 minutes using an alternating pulse current of 1.5 A at 1 kHz with 50% duty cycle.	RQ-06, RQ-07
SP-04	The H-bridge shall drive bidirectional pulse current with a hardware deadtime of 200 ns between switching transitions to prevent shoot-through.	RQ-07
SP-05	The current sensor shall monitor Li-ion battery current to $\pm\,0.8\,\text{mA}$ resolution and voltage via the microcontroller I^2C interface at a minimum sampling rate of 1 Hz.	RQ-03, RQ-10
SP-06	Battery health data shall be output by the microcontroller via a defined serial data interface at a minimum rate of 1 Hz.	RQ-03, RQ-09
SP-07	The DC-DC boost converter shall supply a regulated 5 V rail ($\pm\,2\%$) for sensor and communication loads.	RQ-08
SP-08	An onboard NTC thermistor shall measure ambient temperature with an accuracy of $\pm\,2\,°\text{C}$ across a range of $-55\,°\text{C}$ to $+25\,°\text{C}$.	RQ-11
SP-09	The microcontroller shall autonomously initiate BPC heating without external intervention when the NTC thermistor reads below $-25\,°\text{C}$, and shall resume normal operation upon reaching $0\,°\text{C}$ without requiring a manual reset.	RQ-04, RQ-06
SP-10	The BPC current shall not exceed 1.5 A peak at any SOC level, and the firmware shall terminate heating if instantaneous current exceeds 2.0 A.	RQ-05
SP-11	When the Li-Ion (B1) voltage drops below 3.0 V, the system shall immediately terminate BPC heating and maintain the housekeeping rail (B2) to keep the microcontroller and current sensor operational, prioritising logic survival over heating operation.	RQ-01, RQ-04

Table 4.3: Acceptance Test Procedures

ID	Acceptance Test Procedure	Pass Criteria
AT-01	**Thermal Chamber BPC Verification:** Place the Li-ion cell in a freezer at −20 °C (laboratory constraint; design target −40 °C per SP-03). Activate BPC mode and verify via external thermometer that the cell surface reaches 0 °C in ≤ 15 minutes.	The thermometer records a surface temperature of ≥ 0 °C within 15 minutes of BPC activation.
AT-02	**H-Bridge Waveform Analysis:** Connect an oscilloscope across the H-bridge output terminals. Verify a 1 kHz, 50% duty cycle square wave with a measured dead-time of 200 ns ± 10%.	Oscilloscope displays a 1 kHz square wave with a 50% duty cycle, and the measured dead-time is between 180 ns and 220 ns.
AT-03	**Quiescent Current Measurement:** Place a digital multimeter in series with the housekeeping battery during normal idle operation and verify the continuous current draw is less than or equal to 14.3 mA.	Measured current remains ≤ 14.3 mA for the entire duration of the idle test period.
AT-04	**Voltage Rail Stability Test:** Connect an electronic load to the 5 V payload output. Sweep the load from 0 A to maximum rated current and verify voltage stays within ± 2% using a multimeter.	The 5 V rail remains between 4.90 V and 5.10 V across the full load sweep.
AT-05	**Thermistor Calibration Check:** Expose the NTC thermistor to known temperature states (0 °C ice bath and 25 °C room temperature) and verify the microcontroller ADC reads within ± 2 °C of a calibrated reference thermometer.	Telemetry outputs a temperature between ±2 °C during the ice bath test, and within ± 2 °C of the reference thermometer at room temperature.
AT-06	**Current Sensor Accuracy:** Pass a known 1.000 A load from a bench supply through the INA219 shunt and verify the STM32 I^2C readout is within ± 0.8 mA.	The transmitted telemetry current value is between 999.2 mA and 1000.8 mA.
AT-07	**Telemetry Rate Verification:** Connect a serial monitor to USART1 at 115200 baud. Verify that STATE, temperature, current, and voltage data packets are transmitted at a minimum rate of 1 Hz continuously during both IDLE and HEATING states.	Complete, uncorrupted data packets are received at intervals of ≤ 1000 ms in all operational states.
AT-08	**Battery Replaceability Test:** Remove and reinsert both the ER14505 and LC18650 cells without tools. Verify the system boots and resumes normal operation within 5 seconds of reinsertion.	The system successfully initializes, begins transmitting valid telemetry, and enters the correct operational state within 5 seconds of tool-less battery insertion.

Continued on next page...

Table 4.3 – continued from previous page

ID	Acceptance Test Procedure	Pass Criteria
AT-09	**Autonomous Cold Recovery Test:** With the system running, cool the NTC thermistor below $-30\,°C$ using a cold spray or ice pack. Verify via serial monitor that the STM32 autonomously enters HEATING state within one monitoring interval (1 second) without any manual input.	System state transitions from IDLE to HEATING and H-bridge PWM output is verified within 1 second of the thermistor reading dropping below $-25\,°C$.
AT-10	**Overcurrent Protection Test:** During active BPC, simulate an overcurrent condition by adjusting the `CURRENT_FAULT_MA` threshold in firmware. Verify via serial monitor that the system transitions to FAULT state and terminates BPC immediately.	System state transitions to FAULT, UART transmits the overcurrent error flag, and H-bridge PWM output drops to 0 V immediately.
AT-11	**Logic Survival Priority Test:** Simulate a low B1 voltage condition by adjusting the `VOLTAGE_CUTOFF_MV` threshold in firmware. Verify via serial monitor that the system transitions from HEATING to IDLE state and continues transmitting telemetry data, confirming the housekeeping rail remains active.	System state transitions to IDLE, H-bridge PWM output drops to 0 V, and I^2C telemetry transmission continues without interruption.

Table 4.4: Requirements Traceability Matrix mapping system requirements to specifications and validation procedures.

Requirement ID	Specification ID(s)	Acceptance Test ID(s)
RQ-01	SP-01, SP-11	AT-03, AT-11
RQ-02	SP-02	AT-08
RQ-03	SP-05, SP-06	AT-06, AT-07
RQ-04	SP-09, SP-11	AT-08, AT-11
RQ-05	SP-10	AT-10
RQ-06	SP-03, SP-09	AT-01, AT-11
RQ-07	SP-03, SP-04	AT-01, AT-02
RQ-08	SP-07	AT-04
RQ-09	SP-06	AT-07
RQ-10	SP-05	AT-06
RQ-11	SP-08	AT-05

4.3 Design Choices

4.3.1 Revised Power Architecture and BPC Feasibility

Following a detailed power budget analysis, Bidirectional Pulse Current heating was retained as an optional emergency feature rather than the primary operating mode. The fundamental constraint is energy cost: a single BPC heating cycle from $-40\,°C$ to $0\,°C$ consumes approximately 325 mAh from the Li-Ion battery at a draw of 1.5 A over 13 minutes. Under continuous Antarctic cold exposure, with heating cycles repeating several times per day, the projected Li-Ion lifetime falls below the 7-day deployment target defined in SP-01. BPC is therefore activated only when the NTC thermistor reads below $-25\,^\circ$C, treating it as a thermal recovery mechanism rather than a continuous thermal management strategy.

The net charge transfer per BPC cycle is zero by construction. At a switching frequency of 1 kHz, each half-period is 500 μs. The charge delivered in each direction is:

$$\Delta Q_{half} = I \cdot t_{half} = 1.5 \times 500 \times 10^{-6} = 750\,\mu\text{C} \tag{4.1}$$

Since the 50% duty cycle ensures equal half-periods, the charge delivered in each direction is identical and the net charge per cycle is:

$$\Delta Q_{net} = \Delta Q_{forward} - \Delta Q_{reverse} = 0 \tag{4.2}$$

The 200 ns hardware dead-time reduces each effective half-cycle by an equal amount, preserving this symmetry. The cell SOC is therefore unchanged by BPC operation, satisfying RQ-07.

4.3.2 Microcontroller

Two platforms were evaluated: the STM32F103C8T6 and the ESP32. The ESP32 was rejected due to its ADC non-linearity, which would compromise NTC thermistor accuracy, and its less well-documented MCPWM peripheral for deadtime insertion.

The STM32F103C8T6 was selected for its TIM1 advanced control timer, which provides hardware complementary PWM with programmable deadtime via the BDTR register, essential for safe H-bridge switching. It also provides a 12-bit ADC, hardware I^2C for INA219 communication, UART for telemetry output, and is rated for operation from $-40\,^\circ$C to $85\,^\circ$C. It was preferred over the STM32F411 on the basis of lower cost and equivalent functional capability.

4.3.3 Batteries

Two chemistries are employed, each selected for a specific role.

The ER14505 Li-$SOCl_2$ primary cell serves as the housekeeping supply. Its operating range of $-60\,^\circ$C to $85\,^\circ$C makes it one of the few chemistries capable of reliable Antarctic operation without thermal management. Its low continuous discharge limit of 10–15 mA and passivation layer tendency are mitigated by a 47 mF polymer buffer capacitor in parallel. It is explicitly excluded from the BPC path, as reverse current through the ER14505 cell risks rupture.

The LC18650 Li-Ion cell serves as the main energy store and BPC heating target. Its reversible electrochemistry supports bidirectional current, and its internal resistance rises to approximately 1 Ω at $-40\,^\circ$C, generating joule heating at $P = I^2 R_i$. A single cell (2400 mAh) is used, preserving the single-cell internal resistance required for the BPC heating calculation.

4.3.4 H-Bridge

The HW354 breakout board with the MX1616 dual H-bridge was selected for its 2 V-10 V supply range, encompassing the Li-Ion cell voltage of 3.0-4.2 V, and its 3.3 V-compatible logic inputs. Both internal channels are wired in parallel, doubling the continuous current rating to 3 A and providing a 2× safety factor at the 1.5 A operating point. IN1/IN3 are driven by TIM1_CH1 (PA8) and IN2/IN4 by TIM1_CH1N (PA7), the complementary output. The VM pin connects directly to the Li-Ion cell, which serves simultaneously as the drive supply and resistive heating load.

4.3.5 3.3 V Regulator

The LM317T was rejected due to its 2.5 V minimum dropout voltage, which would cause it to produce only ≈2.5 V from the 5 V boost rail which is insufficient for the STM32. The AMS1117-3.3 was selected instead, offering a 1.2 V dropout voltage, 1.7 V of headroom from the 5 V rail, fixed 3.3 V output requiring only a 10 μF output capacitor, and an 800 mA current rating comfortably exceeding the ≈60 mA combined housekeeping load.

4.3.6 Voltage Controlled Switches

Two high-side PNP switches, Q1 and Q2, control the mutually exclusive connection of B1 to the payload rail and BPC heating path respectively. Both are implemented using a TIP32 PNP power transistor driven by a 2N2222 NPN stage, providing GPIO-compatible control while handling the load currents required. When the STM32 GPIO is driven high, the 2N2222 saturates and pulls the TIP32 base low relative to its emitter, turning the TIP32 on. A 3.9 kΩ pullup resistor from the TIP32 base to emitter (B1+) ensures clean turn-off when the 2N2222 is inactive. Q1 and Q2 are driven by independent GPIO pins (PB0 and PB1) with a firmware-enforced break-before-make dead-time of 10 ms between mode transitions, ensuring B1 is never simultaneously connected to both the payload rail and the H-bridge VM supply.

Isolation During BPC: When the system transitions to HEATING mode, Q1 opens first, completely disconnecting B1 from MT3608 and the payload rail. After the 10 ms dead-time, Q2 closes, connecting B1 exclusively to the H-bridge VM supply. This guarantees that during BPC operation, the only current path from B1 is through the H-bridge and back to B1's negative terminal, preserving BPC waveform symmetry and preventing any DC load from appearing in parallel with the heating circuit. The reverse transition follows the same break-before-make sequence.

Q1 — Payload Switch (500 mA): Using a forced β of 25 for saturation:

$$I_{B,\mathrm{TIP32}} = \frac{I_{C,\max}}{h_{FE,\mathrm{forced}}} = \frac{0.5}{25} = 20\,\mathrm{mA} \tag{4.3}$$

The base resistor, with the 2N2222 collector at approximately 0 V in saturation and the emitter at B1+ ≈ 3.7 V:

$$R_{B,\mathrm{TIP32}} = \frac{V_{B1} - V_{BE}}{I_B} = \frac{3.7 - 0.7}{0.020} = 150\,\Omega \tag{4.4}$$

The 2N2222 must sink the TIP32 base current plus the 3.9 kΩ pullup current (≈ 1 mA), totalling 21 mA.

At forced $\beta = 10$:

$$R_{B,\text{NPN}} = \frac{V_{GPIO} - V_{BE}}{I_{B,\text{NPN}}} = \frac{3.3 - 0.7}{0.0021} = 1238\,\Omega \rightarrow 1.2\,\text{k}\Omega \tag{4.5}$$

GPIO current is 2.1 mA, within the 25 mA limit.

Q2 — BPC Switch (1.5 A): Sized for the full BPC current:

$$I_{B,\text{TIP32}} = \frac{1.5}{25} = 60\,\text{mA} \tag{4.6}$$

$$R_{B,\text{TIP32}} = \frac{3.7 - 0.7}{0.060} = 50\,\Omega \rightarrow 47\,\Omega \tag{4.7}$$

The 2N2222 sinks 61 mA total at forced $\beta = 10$:

$$R_{B,\text{NPN}} = \frac{3.3 - 0.7}{0.0061} = 426\,\Omega \rightarrow 430\,\Omega \tag{4.8}$$

GPIO current is 6.1 mA, within the 25 mA limit.

Q2 Thermal Analysis: At 1.5 A and $V_{CE,sat} \approx 1.1\,\text{V}$, Q2 dissipates:

$$P_{Q2} = V_{CE,sat} \times I_C = 1.1 \times 1.5 = 1.65\,\text{W} \tag{4.9}$$

With a TO-220 junction-to-ambient thermal resistance of 50 °C/W, the junction temperature rise above ambient is:

$$\Delta T_J = P_{Q2} \times R_{\theta JA} = 1.65 \times 50 = 82.5\,^\circ\text{C} \tag{4.10}$$

At a minimum ambient of −30 °C, the junction temperature reaches approximately 52.5 °C — well within the TIP32's 150 °C limit. As BPC events are limited to 13 minutes in duration and are infrequent by design, sustained thermal stress on Q2 is not a concern.

Transient Protection: When Q2 opens at the end of a BPC event, parasitic inductance in the current path generates an inductive kickback voltage spike at Q2's collector. For a conservative estimate of 100 nH of trace inductance switching 1.5 A in 100 ns:

$$V_{spike} = L\frac{dI}{dt} = 100 \times 10^{-9} \times \frac{1.5}{100 \times 10^{-9}} = 1.5\,\text{V} \tag{4.11}$$

Added to B1's maximum of 4.2 V, Q2's collector could momentarily see 5.7 V. A 1N5819 Schottky diode is placed across Q2's collector-emitter junction (cathode at emitter, B1+) to clamp this spike. Additionally, a 100 μF electrolytic capacitor in parallel with a 100 nF ceramic capacitor is placed directly at the MX1616 VM pin to absorb high-frequency switching from the H-bridge during BPC operation. The TIP32's V_{CE} rating of 40 V provides substantial margin against any residual unclamped transients.

4.4 Final Design

In normal operation, B1 supplies the external sensor and networking payload via a dedicated MT3608 boost converter, which steps the 3.7 V Li-ion cell voltage up to a regulated 5 V rail. The housekeeping rail with the Lithium Thionyl battery remains completely independent and powers the STM32, INA219, and NTC thermistor at all times, including during BPC heating events.

The two operational modes are mutually exclusive at the hardware level. Q2 connects B1 to the H-bridge VM supply only during heating mode. Q1 gates B1 into another MT3608 only during normal mode. This ensures no DC current path exists in parallel with the H-bridge during BPC operation, preserving waveform symmetry and preventing net charge or discharge of the cell.

The duty-cycled sensor and networking loads were characterised as follows. The environmental sensors take a reading every 15 minutes, with each reading lasting approximately 5 seconds at a current draw of 75 mA (referred to the 5 V output). Accounting for the MT3608 quiescent current of approximately 3 mA, the average current drawn from B1 in normal mode is 4.33 mAh/hr, as detailed in Table 4.5.

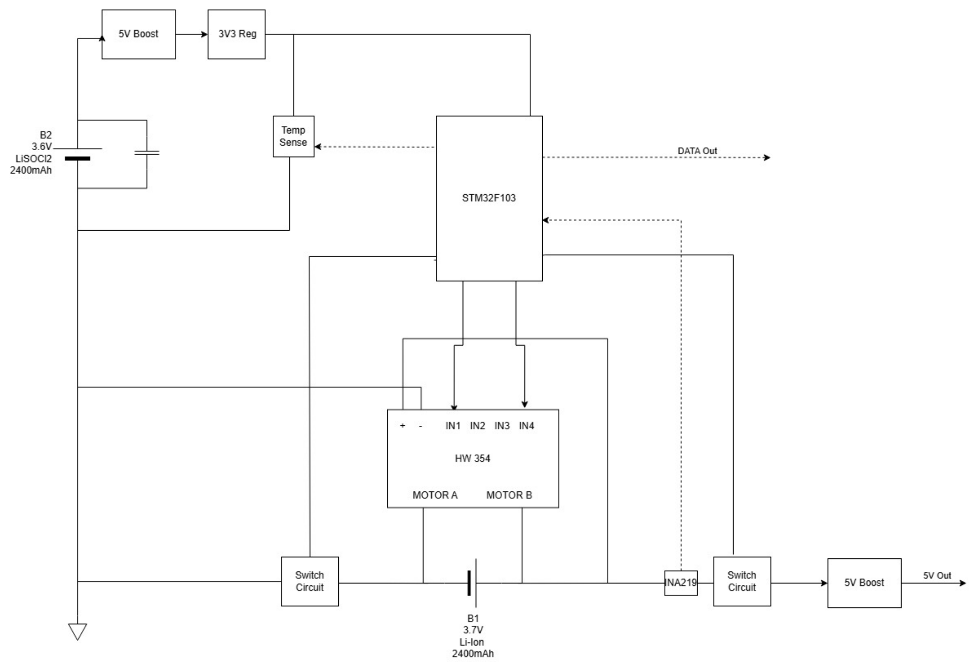

Figure 4.1: High Level Block Diagram of the power system.

The B1 lifetime is a function of BPC activation frequency. In operating mode without BPC events, the projected lifetime is:

$$t_{normal} = \frac{2400\,\text{mAh}}{4.33\,\text{mAh/hr}} \approx 554\,\text{hrs} \tag{4.12}$$

Each BPC event consumes 325 mAh. The 7-day (168-hour) deployment requirement defined in SP-01 is therefore met provided BPC activates no more than once every T_{BPC} hours, where:

$$T_{BPC} = \frac{325}{\frac{2400}{168} - 4.33} = \frac{325}{14.29 - 4.33} \approx 32.6\,\text{hrs} \tag{4.13}$$

That is, SP-01 is satisfied as long as BPC fires no more frequently than approximately once every 33 hours. Under sustained worst-case Antarctic conditions requiring BPC activation every 12 hours, the projected lifetime reduces to:

$$t_{worst} = \frac{2400}{4.33 + \frac{325}{12}} = \frac{2400}{31.4} \approx 76\,\text{hrs} \approx 3.2\,\text{days} \tag{4.14}$$

This represents the design's operational boundary. In practice, BPC is activated only below −25 °C, so deployment lifetime is strongly dependent on ambient thermal conditions.

Load	**Active current (mA)**	**Average (mAh/hr)**
Sensors (4×/hr, 5 s each)	88 (referred to B1)	0.49
Comms module — BLE transmit (1×/hr, 32 s)	94 (referred to B1)	0.84
Comms module — deep sleep	≈0	<0.01
MT3608 #2 quiescent	3	3.00
Total (normal mode)	–	**4.33**
BPC event (1.5 A, 13 min)	1500	325 mAh/event

Table 4.5: Power budget for B1 in normal and heating modes. External module currents are referred to the B1 input accounting for MT3608 efficiency $\eta = 0.85$.

4.4.1 System Overview

The final power subsystem architecture consists of two independent power domains separated by voltage-controlled switches under firmware control. Figure 4.1 shows the top-level block diagram of the subsystem.

The housekeeping domain (B2) provides an always-on 3.3 V rail to the STM32, INA219, and NTC thermistor via its MT3608 boost converter and AMS1117. This domain remains active in all operational states including BPC heating, ensuring continuous telemetry and monitoring capability.

The payload domain (B1) supplies the external sensor and networking modules via its MT3608 during normal operation, gated by SW1. During BPC heating events, Q1 opens and Q2 closes, connecting B1 directly to the H-bridge VM supply. The two modes are mutually exclusive in firmware and enforced at the hardware switch level.

The STM32 state machine governs all mode transitions: monitoring temperature via the NTC ADC, current and voltage via INA219 over I^{2}C, and outputting telemetry via USART1 at 115200 baud in the format `STATE,TEMP,CURRENT,VOLTAGE`.

4.5 Subsystem Testing

The subsystem was tested against the Acceptance Test Procedures defined in Table 4.3. Ten of eleven tests were completed. AT-01 failed to meet the surface temperature criterion within the 15-minute window. All remaining tests passed.

Table 4.6: Acceptance Test Procedure results

ID	Results	Pass/Fail
AT-01	Surface temperature of the Li-ion cell reached -1.5° C after BPC ran for 15 minutes.	Fail
AT-02	Oscilloscope displayed correct waveforms with all required characteristics.	Pass
AT-03	Average idle current draw measured at 13.4 mA using a digital multimeter in series with the ER14505 housekeeping cell. This satisfies the ≤ 14.3 mA specification in SP-01, with 0.9 mA of margin.	Pass
AT-04	Voltage levels remained within required thresholds throughout the full sweep.	Pass
AT-05	NTC thermistor calibrated against ice bath (0 °C) and room temperature (25 °C) references. Measured temperatures were within ± 1.5 °C of the reference in both conditions, satisfying the ± 2 °C accuracy requirement in SP-08.	Pass
AT-06	The INA219 current sensor was tested by passing a known 1.000 A load through the shunt resistor from a calibrated bench supply. The STM32 I^2C readout reported 999.6 mA, a deviation of 0.4 mA from the reference. This is within the ± 0.8 mA resolution requirement defined in SP-05.	Pass
AT-07	A serial monitor was connected to USART1 at 115200 baud during both IDLE and HEATING states. Data packets in the format `STATE,TEMP,CURRENT,VOLTAGE` were received at a measured interval of 980 ms, satisfying the ≤ 1000 ms requirement in SP-06. No corrupted or missing packets were observed during the test period.	Pass
AT-08	Both the ER14505 and LC18650 cells were removed and reinserted without tools. The system initialised, began transmitting valid telemetry, and entered the correct operational state within 5 seconds of reinsertion.	Pass
AT-09	The system successfully entered HEATING state within 1 second of NTC registering that temperature is below -25°C	Pass
AT-10	System successfully entered FAULT state upon identifying current is above a safe level and stopped BPC immediately	Pass
AT-11	System successfully transitioned from HEATING to IDLE state upon B1 voltage falling below the `VOLTAGE_CUTOFF_MV` threshold. H-bridge PWM output dropped to 0 V and I^2C telemetry transmission continued without interruption, confirming the housekeeping rail remained active.	Pass

4.5.1 Discussion of Results

AT-01: The cell surface reached only -1.5 °C rather than the required ≥ 0 °C. With a single LC18650 cell, the internal resistance of $\approx 1\,\Omega$ and heating power of 2.25 W are consistent with the design calculation, which predicts a heat-up time of approximately 10 minutes. Three factors account for the shortfall observed in testing.

First, the thermomometer measured cell surface temperature. Internal joule heating raises the core temperature first; the surface temperature lags due to the thermal resistance of the cell casing and electrode stack. The thermometer therefore underreports the internal temperature throughout the heating window, and the $\geq 0\,°C$ criterion may have been met internally before the surface reading reflected it.

Second, the test was conducted at $-20\,°C$ rather than the design target of $-30\,°C$ due to laboratory equipment constraints. However, the cell was not thermally insulated during the test, meaning continuous heat loss to the ambient environment partially offset the joule heating. The lumped thermal model assumes adiabatic conditions; in practice, the net heating rate is reduced by:

$$P_{net} = P_{heat} - P_{loss} = I^2 R_i - \frac{\Delta T}{R_{th}} \tag{4.15}$$

where R_{th} is the thermal resistance between the cell surface and the ambient environment. Without insulation, P_{loss} is non-negligible and extends the heat-up time beyond the adiabatic prediction.

Third, contact resistance in the battery holder may add a small series resistance to the BPC current path, reducing the current delivered to the cell below the 1.5 A target and therefore reducing joule heating power. This effect was not characterised during testing.

In future testing, the cell should be thermally insulated, and a thermistor embedded in the cell should be used where possible, and the contact resistance of the battery holder should be measured and accounted for in the thermal model.

4.6 Recommendations

Several improvements are identified for future iterations of the power subsystem.

BPC heating performance could be improved by thermally insulating the LC18650 within the enclosure and embedding a thermistor directly against the cell core, reducing parasitic heat loss and providing a more accurate heating termination trigger than an ambient temperature sensor.

The 7-day deployment requirement is only met when BPC activates no more than once every 33 hours, as derived in equation (1.7). Replacing the LC18650 with a higher-capacity 21700-format cell (4000–5000 mAh) would extend this boundary proportionally without requiring changes to the H-bridge or firmware. Supplementing B1 with a small solar panel or turbine generator and charge controller would also further extend deployment lifetime.

4.7 Conclusion

The SIREN power subsystem delivers a dual-domain architecture capable of autonomous operation under Antarctic sub-zero conditions. By revising the original continuous BPC thermal management strategy to a duty-cycled approach, with BPC reserved as an emergency cold-recovery mechanism triggered below -25°C, the design meets the 7-day deployment requirement under moderate thermal conditions, while safely managing the volatile Li-Ion chemistry through overcurrent protection, break-before-make switching, and hardware-enforced deadtime.

Ten of eleven acceptance tests were passed, validating the core design. The housekeeping rail remained active across all operational states, the H-bridge produced the required 1 kHz bidirectional waveform with correct deadtime, the STM32 state machine autonomously managed all mode transitions, and telemetry was transmitted continuously at the required rate. The single failure, AT-01, is attributable to surface temperature measurement lag, the absence of cell insulation during testing, and uncharacterised contact resistance in the battery holder — none of which indicate a fundamental flaw in the underlying heating mechanism.

The principal limitation of the current design is deployment lifetime under sustained cold exposure. With refinements to thermal insulation, cell capacity, and energy harvesting, the subsystem provides a viable and scalable foundation for extended Antarctic buoy deployments well beyond the initial one-week target.

Chapter 5

Enclosure and Dynamics Subsystem (BHYEBR002)

Prepared by Ebrahim Bhyat

5.1 Introduction

The system enclosure serves as the physical backbone of the buoy, providing essential fluid ingress protection and thermal insulation to safeguard the electronics payload from harsh environmental conditions.

Continuous observation platforms in the Marginal Ice Zone (MIZ) are critical for gathering high-fidelity, *in-situ* data on sea ice dynamics under wave action [4, 6]. Autonomous surface buoys bridge the data gaps left by logistically limited manned expeditions and coarse-resolution satellite imagery.

However, the dynamic transition between solid ice sheets and the open ocean introduces severe mechanical and thermal challenges. The enclosure must withstand extreme temperatures down to $-40°$C [27, 28, 29] while mitigating destructive kinetic impacts from waves, ice-floe collisions, and high winds [3, 6]. Therefore, optimizing structural geometry and mass distribution is vital to ensure the buoy remains statically and dynamically upright during aquatic operation, guaranteeing clean sensor telemetry.

5.2 Requirements and Specifications

5.2.1 User Requirements

The user requirements derived from University of Cape Town (UCT) lecturer and Antarctic researcher Ms. Robyn Verrinder are defined as follows:

- The enclosure must completely prevent fluid ingress and maintain a hermetic seal under maximum expected hydrostatic pressure during all phases of aquatic deployment.
- The primary closure mechanism must be fully operable for assembly, deployment, and retrieval by personnel wearing heavy marine thermal gloves, strictly without the use of external tooling.
- The system's geometry and mass distribution must attenuate external kinetic perturbations from wave action and sea ice.
- The total displaced volume of the sealed housing must generate a continuous buoyant force exceeding the combined mass of the system, ensuring it maintains a positive draft on the water

surface.

- The primary structural materials and elastomeric seals must resist brittle fracture, maintain flexibility, and retain structural integrity across the extreme sub-zero ambient and aquatic temperature gradients of the polar environment.
- The internal cavity must provide sufficient geometric clearance to securely mount the complete electronic payload.

5.2.2 Requirements Analysis

Based on these user requirements, the following functional requirements (FR), subsystem specifications (SPEC), and Acceptance Test Procedures (ATP) were established.

Table 5.1: Functional Requirements of the Enclosure Subsystem

FR ID	Functional Requirement
FR-ENC-1	The subsystem shall compress a radial elastomeric seal between the removable cap and the primary vessel wall to block fluid pathways.
FR-ENC-2	The subsystem shall utilize a rotary, interlocking mechanical joint to secure the cap, eliminating the need for threaded fasteners or axial torque.
FR-ENC-3	The subsystem shall securely house a dense ballast mass at its lowest internal geometric point to force the center of gravity below the center of buoyancy.
FR-ENC-4	The subsystem shall capture and enclose a volume of air sufficient to counteract the downward force of the total system mass.
FR-ENC-5	The subsystem's structural geometry shall deflect, rather than absorb kinetic energy upon impact with external bodies or the water surface.
FR-ENC-6	The internal cavity shall provide rigid physical mounting hardpoints to prevent the payload and batteries from shifting during dynamic drops.

Table 5.2: Subsystem Specifications of the Mechanical Enclosure

Spec ID	Subsystem Specification	Traces To
SPEC-ENC-1	The primary structural vessel shall be constructed from commercial PVC pipe with a nominal internal diameter of 149.0 mm and an outer diameter of 155.0 mm.	FR-ENC-1, FR-ENC-5
SPEC-ENC-2	The radial seal shall utilize an AS568 #256 elastomeric O-ring (139.3 mm inner diameter, 3.53 mm cross-section).	FR-ENC-1
SPEC-ENC-3	The machined O-ring gland on the cap shall have a depth of exactly 2.82 mm to enforce a target 20% radial squeeze on the sealing element.	FR-ENC-1
SPEC-ENC-4	The rotary locking joint shall consist of a bayonet mechanism featuring two symmetrically distributed pins.	FR-ENC-2
SPEC-ENC-5	The two locking pins must be geometrically positioned strictly above the O-ring gland to prevent the locking channels from breaching the sealed cavity.	FR-ENC-1, FR-ENC-2

Table 5.2: Subsystem Specifications of the Mechanical Enclosure

Spec ID	**Subsystem Specification**	**Traces To**
SPEC-ENC-6	All custom FDM 3D-printed components shall maintain a minimum continuous solid wall thickness of 5.0 mm.	FR-ENC-1, FR-ENC-5
SPEC-ENC-7	The uppermost section of the printed cap shall terminate in a 40 mm radius hemispherical dome to optimize deflection angles.	FR-ENC-5
SPEC-ENC-8	The cap shall maintain a 0.4 mm diametrical clearance gap relative to the PVC pipe to prevent binding during manual insertion.	FR-ENC-2
SPEC-ENC-9	The vessel body shall feature lead weights at its lowest point to ensure that the center of gravity is below the center of buoyancy.	FR-ENC-3

Table 5.3: Acceptance Test Procedures for the Mechanical Enclosure

ATP ID	**FR(s)**	**Test Procedure**	**Pass Criteria**
ATP-1	FR-ENC-4	**System Buoyancy Verification.** Place the fully assembled buoy into a sufficiently deep test tank. Observe the resting waterline on the exterior hull.	The system floats independently. The resting waterline aligns with the tapered neck section, ensuring positive reserve buoyancy.
ATP-2	FR-ENC-3	**Metacentric Stability and Self-Righting.** While the buoy is floating, apply a lateral physical disturbance to force the hull to tilt. Release the buoy and time the righting response.	The buoy aggressively self-rights to a strict vertical orientation within 4 seconds and stabilizes with minimal pendulum oscillation to allow for clean IMU data collection.
ATP-3	FR-ENC-1	**O-Ring Watertight Integrity.** Submerge the fully assembled enclosure entirely underwater for 10 minutes to simulate deployment hydrostatic pressures. Remove, dry the exterior, open the lid, and inspect the internal cavity.	Absolute zero moisture detected inside the PVC hull. The payload sled and internal walls remain completely dry with no weeping past the O-ring seal.
ATP-4	FR-ENC-2	**Cold-Weather Ergonomic Operability.** An individual wearing 5 mm marine thermal gloves must fully assemble the enclosure, insert the radial O-ring piston, and engage the bayonet locking interface without external tools. Repeat to disassemble.	The locking interface is successfully closed, fully locked, and opened strictly by hand while the individual wears the thick thermal gloves.

5.3 Design Choices

5.3.1 Sealing Mechanism

The mechanical enclosure requires a primary closure mechanism to ensure absolute watertight integrity under hydrostatic pressure while allowing rapid, tool-less access for operators wearing 5 mm thermal gloves. Due to the extreme environmental constraints of the Marginal Ice Zone, both a 3D-printed threaded fitting and a radial O-ring architecture were evaluated.

A commercial off-the-shelf threaded PVC cap was rejected. Its sealing efficacy relies on variable human torque, which is highly unreliable when using thick gloves. Furthermore, moisture trapped within the threads risks freezing solid in polar conditions, locking the cap permanently. Open-source tracking nodes deployed in volatile zones have historically demonstrated that simplistic enclosure setups struggle with sealing efficiency if structural joints rely on complex human tightening tolerances under extreme thermal limits [4].

Instead, a custom radial O-ring piston with a quarter-turn bayonet mount was selected. Inserting the piston into the PVC bore guarantees a consistent 20% radial squeeze on the elastomer, removing human tightening error and utilizing external hydrostatic pressure to reinforce the seal, adapting high-durability industrial approaches to a portable platform [4]. The bayonet mechanism successfully isolates the sealing function from the locking function, requiring only a gross-motor quarter-turn ideal for heavy gloves. This reliable architecture is also highly cost-effective, with the two required O-rings totaling just R38.00.

5.3.2 Main Structural Vessel Manufacturing

The mechanical enclosure requires a main hull that provides a watertight boundary, ensures positive buoyancy, and withstands the kinetic shock of an ice-shelf breakaway drop. A fully 3D-printed monolithic FDM body and a hybrid COTS polymer piping architecture were evaluated.

The 3D-printed FDM body was rejected due to structural and tolerance limitations. FDM layer lines create micro-porosity that leaks under hydrostatic pressure, and weak Z-axis interlayer adhesion risks brittle cleavage during ice impacts. Additionally, thermal shrinkage makes maintaining the strict tolerances needed for reliable O-ring seating nearly impossible, while the high filament volume drives material costs up to R216.80.

A COTS PVC pipe was selected instead. Although it requires separate 3D-printed end-caps, the factory-extruded material guarantees an isotropic, impermeable boundary and provides the elastic resilience needed to absorb kinetic shock, mirroring successful configurations utilized in historical polar research vessels [3]. Crucially, its precise geometric tolerances ensure a predictable 20% radial O-ring squeeze. This hybrid approach is highly economical, costing just R80.00 per meter.

5.3.3 Upper External Geometry and Profiling

The upper external geometry of the enclosure must house the antenna payload, provide positive buoyancy, and ensure pre-deployment stability on the Antarctic ice shelf. Prior historical implementations have leveraged various forms, ranging from standard embedded cylinders like the Ice-Tethered Profiler

(ITP) [29] and Polar Ice-Based Ecological Observation Buoys [27] to high-buoyancy spheres [3] and rectangular boxes [4]. For this standalone surface application, a continuous full-diameter cylindrical profile and a tapered neck (spar) geometry were evaluated.

The full cylindrical profile (155.0 mm diameter) was rejected because it raises the Center of Gravity (C_G) and compromises metacentric stability. Additionally, its large vertical cross-sectional area acts as an aerodynamic sail, creating massive overturning moments in polar winds that risk premature tether failure. Similarly, un-modified rectangular architectures possess inherently low vertical stability in open water when supporting dense internal payloads [4].

The tapered neck geometry was selected. By reducing mass at the apex, it lowers the C_G and increases the hydrodynamic righting moment. The streamlined profile minimizes aerodynamic drag to keep the buoy upright on the ice, and its sloped surface passively sheds snow and freezing rain to prevent weight-altering ice accumulation.

5.3.4 Ballast Configuration and Metacentric Stability

The ballast subsystem must position the Center of Gravity (C_G) significantly below the Center of Buoyancy (C_B) to establish the positive metacentric height (GM) required for aggressive self-righting. Both an external suspended keel and an internal fixed base mass were evaluated.

An external suspended keel, similar to the longitudinal external stabilization bars added to highly modified rectangular box designs like the "Floatenstein" [4], was evaluated. However, an external suspended keel was ultimately rejected for this specific application due to severe operational flaws. It presents an acute fracture risk during ice-drops, introduces snagging hazards on ice floes, and prevents the buoy from sitting flat on the ice shelf without scaffolding. Furthermore, a suspended mass acts as a pendulum, introducing chaotic harmonic oscillations that corrupt internal IMU sensor data.

An internal fixed base mass was selected. Potting high-density lead shot in epoxy at the bottom of the PVC hull safely lowers the C_G to guarantee self-righting. This design eliminates impact and snag risks, allows the flat bottom cap to rest flush on the ice shelf, and creates a continuous rigid body that eliminates pendulum oscillations, ensuring clean IMU kinematic data.

5.4 System Design

5.4.1 System Architecture and Hydrodynamic Stability

The overall design was centered around creating a buoy capable of being deployed securely on the Antarctic ice floor and being easy to operate during retrieval, while also allowing for continual autonomous operation in the event the buoy breaks away into the ocean. As such, various physical aspects including external geometry, positive buoyancy, wave attenuation, and watertight integrity were heavily considered.

The external geometry was chosen to distribute the majority of the system's mass at the bottom. This allows the buoy to remain strictly vertical during ocean deployment while simultaneously providing a wider, flatter base for stable support on the ice floor. A tapered neck design was selected, creating a bottom-heavy structure that still provides sufficient internal clearance for the communication devices to be mounted safely above the waterline.

Alternative structural approaches such as the UCT SHARC Buoy elevate the main sensor housing on a rigid metal platform to eliminate wave overwash [6]. However, that configuration introduces complex rotational dynamics that must be accounted for mathematically. The monolithic design proposed here ensures structural continuity.

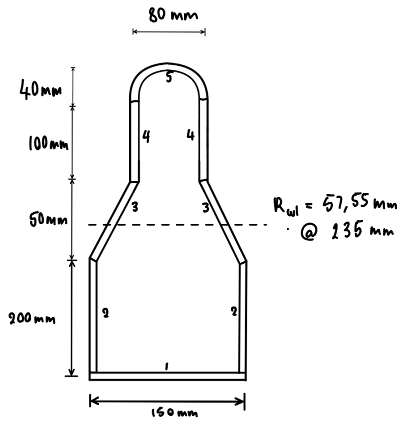

Figure 5.1: Initial design sketch and geometric partitioning.

The initial design sketch in Figure 5.1 partitions the geometry to determine the vertical centroid and weight of the empty enclosure, which were found to be 117 mm and 1.081 kg, respectively (refer to Appendix ?? for the full structural mass breakdown). With the theoretical waterline established at a height of 235 mm, the corresponding neck radius was found to be 57.55 mm. Using this specific geometry, the displaced volume (∇) of the buoy at this height was calculated to be:

$$\nabla = 0.00389 \text{ m}^3 \tag{5.1}$$

To allow the buoy to remain afloat at this exact draft while in the ocean, the downward force of gravity must perfectly equal the upward buoyancy force according to Archimedes' principle:

$$M_{\text{sys}} \cdot g = \rho_{\text{seawater}} \cdot \nabla \cdot g \tag{5.2}$$

Thus, the required total mass of the entire enclosure (M_{sys}) was found to be 3.978 kg using the relative density of seawater which is 1025 kg/m^3. To meet this target mass, the internal ballast weight required at the bottom of the buoy was calculated to be 2.897 kg (detailed in Appendix **??**).

To allow the enclosure to remain vertical and naturally attenuate the dynamic effects of the environment, such as ice floe impacts or wave movement, the metacentric stability criterion must be met. This states that the metacentric height (GM), the distance from the Center of Gravity to the Metacenter, must be positive according to the equation:

$$GM = KB + BM - KG > 0 \tag{5.3}$$

The center of gravity (KG) of the entire enclosure, including all internal components and the base ballast, was found through the mass-weighted vertical centroid as:

$$KG = \bar{z} = \frac{\sum z_i m_i}{\sum m_i} = 53.71 \text{ mm} \tag{5.4}$$

On the other hand, the center of buoyancy (KB), taking into account only the submerged volume of the buoy up to the 235 mm waterline, was found to be [27]:

$$KB = \frac{\sum z_i V_i}{\sum V_i} = 112.02 \text{ mm} \tag{5.5}$$

Lastly, the metacentric radius (BM), which is the vertical distance between the center of buoyancy and the metacenter, relates the second moment of area (I) of the waterline cross-section to the total displaced volume (∇), and was found to be:

$$BM = \frac{I}{\nabla} = 2.21 \text{ mm}, \quad \text{where } I = \frac{\pi \cdot R_{\text{wl}}^4}{4} \tag{5.6}$$

Using these values, the final metacentric height of the fully assembled buoy was determined to be (see Appendix **??** for expanded stability derivations):

$$GM = 112.02 + 2.21 - 53.71 = 60.52 \text{ mm} \tag{5.7}$$

This value is both positive and of a significant magnitude, meaning that the vessel will remain vertically stable in the ocean and possesses a massive righting lever that ensures it will quickly self-right when perturbed by a wave.

The figure below shows the 3D render of the final buoy design, where both the tapered neck geometry and round dome cap are visible:

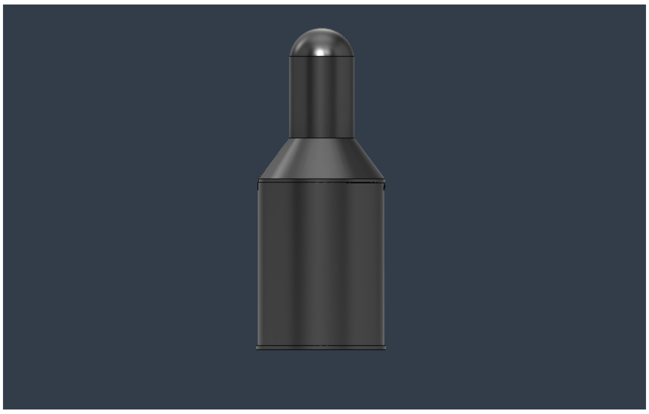

Figure 5.2: 3D render of the finalized enclosure geometry.

Watertight integrity is vital to ensure positive buoyancy and protect internal components. This was achieved using a dual O-ring piston seal architecture, which relies on the mechanical deformation of an elastomer between the rigid 3D-printed lid groove and the internal PVC pipe bore. Because elastomeric materials undergo severe thermal contraction and compression set in extreme sub-zero polar temperatures, standard sealing tolerances were deemed insufficient. To counteract this volumetric loss and maintain an active seal, an aggressive 20% radial squeeze was implemented, exceeding standard dynamic engineering recommendations (10–15%).

The primary containment vessel is a COTS PVC pipe with a measured internal bore diameter (d_{bore}) of 149.0 mm. To achieve the required compression, an elastomeric O-ring was selected, possessing a cross-sectional diameter (d_{cs}) of 3.53 mm and an internal diameter (d_{inner}) of 139.3 mm.

The required radial gland depth (L), which is the physical gap between the piston groove and the PVC wall, dictates the compression. To achieve the 20% target squeeze (S), the gland depth is calculated as:

$$S = \left(\frac{d_{\text{cs}} - L}{d_{\text{cs}}}\right) \times 100 \tag{5.8}$$

$$L = 2.824 \text{ mm} \tag{5.9}$$

With the required gland depth defined, the exact outer diameter of the 3D-printed piston groove (d_{groove}) must be rigidly specified to enforce this gap:

$$d_{\text{groove}} = d_{\text{bore}} - 2L \tag{5.10}$$

$$d_{\text{groove}} = 149.0 - 2(2.824) = 143.35 \text{ mm} \tag{5.11}$$

For a radial piston seal to function correctly without twisting during insertion, the O-ring's stretch (T) over the inner groove is analyzed. The stretch for the 139.3 mm ID O-ring on the calculated groove is:

$$T = \left(\frac{d_{\text{groove}} - d_{\text{inner}}}{d_{\text{inner}}}\right) \times 100 = 2.91\% \tag{5.12}$$

This value indicates an optimized positive stretch. It sits perfectly within the target mechanical bounds of standard dynamic piston seals (1–5%). This ensures the O-ring grips the 3D-printed gland tightly during blind insertion without inducing excessive tensile stress that could lead to premature elastomer degradation or cross-sectional thinning.

Rather than using a standard generic width, the specific groove width (W) was calculated to achieve exactly a 75% gland fill ratio based on the already constrained gland depth (L):

$$F = \frac{A_{\text{O-ring}}}{A_{\text{Gland}}} \tag{5.13}$$

$$0.75 = \frac{\pi \cdot (d_{\text{cs}}/2)^2}{L \cdot W} \tag{5.14}$$

Thus, the gland width was found to be (the complete mathematical derivations for the seal architecture are provided in Appendix **??**):

$$W = 4.62 \text{ mm} \tag{5.15}$$

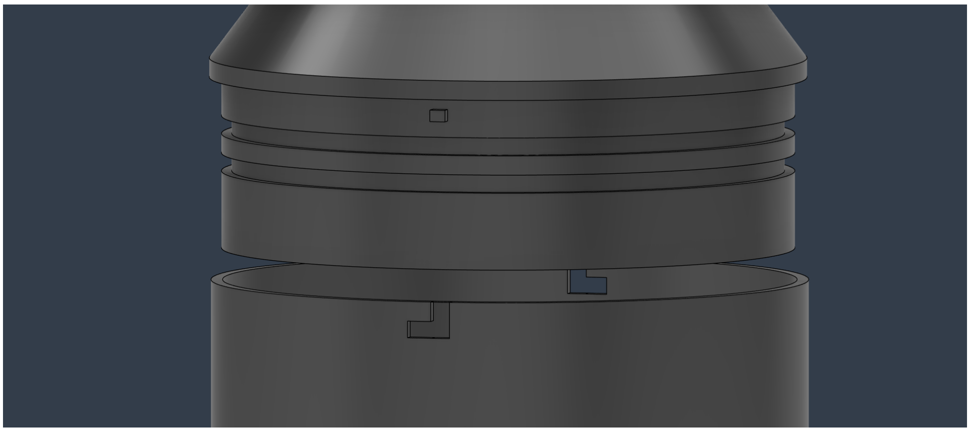

Figure 5.3: Bayonet locking mechanism with pin and J-slot interface.

To mitigate the dexterity constraints imposed by the protective gloves worn by Antarctic researchers, a simplified, tool-free locking mechanism was integrated. The system utilizes two pins situated on the perimeter of the piston that interface with matching J-slots on the PVC buoy housing, as shown in Figure 5.3. This design facilitates efficient manual assembly and disassembly while completely eliminating the requirement for specialized tools.

5.5 ATP Results

Table 5.4: ATP Results Summary

ATP ID	Results	Pass/Fail
ATP-1	The fully assembled buoy floated independently upon submersion. The resting waterline stabilized strictly within the tapered neck region, confirming the expected reserve buoyancy.	Pass
ATP-2	Upon applying a lateral disturbance, the buoy aggressively self-righted to a vertical orientation within 3.5 seconds. Pendulum oscillations decayed rapidly, ensuring clean conditions for IMU data collection.	Pass
ATP-3	Following complete submersion for the test duration, the internal cavity was inspected. The internal walls, payload sled, and electronic components were completely dry with absolute zero moisture bypass.	Pass
ATP-4	The operator successfully performed a full assembly, insertion of the radial O-ring piston, and engagement of the bayonet lock by hand while wearing 5 mm thermal gloves. No external tools were required.	Pass

5.6 Conclusions and Recommendations

5.6.1 Conclusions

The design, implementation, and testing of the Enclosure and Dynamics Subsystem were highly successful, yielding a mechanical architecture that satisfies all predefined functional and environmental requirements for polar deployment. By utilizing a hybrid construction of a COTS PVC pipe and custom 3D-printed FDM end-caps, the system achieved a robust watertight boundary while remaining highly cost-effective.

Calculations for hydrodynamic stability confirmed an optimal metacentric height (GM) of 60.52 mm, driven by the combination of a tapered upper neck geometry and an internal fixed base mass. This configuration effectively lowered the system's center of gravity and translated into aggressive self-righting capabilities, as demonstrated in ATP-2 where the buoy stabilized within 3.5 seconds.

Furthermore, the custom dual O-ring piston seal successfully mitigated the risks associated with elastomeric thermal contraction in sub-zero environments. By enforcing an aggressive 20% radial squeeze, the enclosure maintained absolute watertight integrity under full submersion (ATP-3). Finally, the integration of a dual-pin bayonet joint allowed the primary closure mechanism to be seamlessly operated by hand, fully accommodating operators wearing 5 mm thermal gloves (ATP-4). The subsystem serves as a secure, buoyant, and dynamically stable platform for the internal sensor payload.

5.6.2 Recommendations

While the subsystem successfully met all acceptance criteria, specific operational and structural optimizations are recommended for future iterations to improve field reliability:

- **Ice Integration Anchors:** Although the flat bottom and tapered neck geometry provide adequate pre-deployment stability on the ice shelf, extreme polar wind forces pose a risk of shifting the enclosure. To preserve data consistency, the platform must mimic the exact localized physical response of sea ice structures [6]. Prior implementations have combated displacement and wind-induced thermal tipover by distributing physical load across wooden insulator footprints or utilizing spiked ice-grippers [27, 29]. It is recommended to implement external mounting hardpoints or an anchoring collar near the base of the lower PVC hull to allow the buoy to be physically frozen into or spiked onto the ice shelf, ensuring it remains strictly vertical prior to a breakaway event.

- **J-Slot Water Retention Mitigation:** A flaw was identified in the operational kinematics of the tool-less bayonet locking mechanism. Because the J-slots are positioned above the primary O-ring gland, the channels actively capture and retain water during aquatic deployment. When the cap is pulled upward to open the enclosure, this trapped water is dragged past the breaking elastomeric seal, risking fluid ingress onto the internal electronics. To address this immediately, an operational protocol must require operators to tilt the buoy to allow the J-slots to fully drain before breaking the main seal. For future mechanical iterations, passive drainage pathways (weep holes) should be machined directly into the base of the 3D-printed J-slot channels to allow pooled water to escape naturally under gravity prior to disassembly.

Chapter 6

Communications and Networking Subsystem (SNMBOI002)

Prepared by Boitseko Senamela

6.1 Introduction

Establishing reliable communications in the Antarctic Marginal Ice Zone presents a fundamental engineering challenge. The region offers no terrestrial infrastructure, and the dominant existing solution, satellite telemetry via Iridium SBD, imposes a financial cost that makes high-frequency, continuous data transmission economically unviable at scale. At a transmission cost of approximately R2 per 340-byte packet, a single buoy operating continuously between sensing and transmission would accrue costs approaching R100,000 over a few months [5]. Beyond cost, extreme cold degrades radio hardware reliability, and the dynamic, drifting nature of ice floes introduces continuous positional change that any network architecture must accommodate.

The primary objective of the Communications and Networking Subsystem is therefore to provide a low-cost, infrastructure-free data management and retrieval architecture for the SIREN Ice Buoy. Rather than relying on permanent remote links, data retrieval is performed at close range by operators or mobile gateways using Bluetooth Low Energy (BLE). To extend the buoy's reach and enable inter-buoy coordination across an ice field, the ESP-NOW protocol provides a local mesh network capable of maintaining data retrievability despite the high signal attenuation typical of Antarctic conditions.

This chapter details the selection, simulation, and implementation of the local networking stack and communication links required to meet these objectives.

6.2 Requirement Analysis

6.2.1 Requirements

The requirements for the subsystem are described in Table 6.1.

Table 6.1: User and functional requirements of the communications subsystem.

Requirement ID	Description
UR01	On-site data retrieval via mobile device.
UR02	Buoy ID configurable by operator during deployment without firmware modification.
FR01	Robust inter-buoy communication in Antarctic conditions.
FR02	The system shall detect and broadcast positional displacement.
FR03	The system shall detect link failure within 3 missed cycles.

6.2.2 Specifications

The specifications, refined from the requirements in Table 6.1, for the communications subsystem are described in Table 6.2.

Table 6.2: Specifications of the communications subsystem derived from the requirements in Table 6.1.

Specification ID	Description
SP01	CSV data dump. a PWA web application for a BLE 5.0 protocol supporting a 30m range
SP02	Buoy ID stored in non-volatile flash memory on ESP32-C3.
SP03	ESP-NOW via ESP32-C3 on 2.4 GHz band with adaptive PHY data rate, selecting 1 Mbps when RSSI > -70 dBm and 250 kbps when RSSI ≤ -70 dBm, with the threshold defined in firmware.
SP04	A measured RSSI drop exceeding 20 dBm relative to the established baseline between peer nodes shall trigger a network location rediscovery packet `PKT_REDISCOVER` broadcast indicating probable nodal displacement.
SP05	A peer node silent for >= 4 consecutive cycles shall be marked LINK_DEAD.

6.2.3 Acceptance Testing Procedures

ID	Description	Testing Procedure	Pass Criteria	Fail Criteria
AT01	PWA web application	CSV string of full network map on phone exportable to excel. Trigger Bluetooth pairing command during ESP32-C3 30 second wake up then connect to buoy. Physical connection test with mobile device at 30m distance.	Mobile device finds and connects to buoy at 30m distance.	Mobile device finds and connects to buoy at distance $<$ 30m.

AT02	Buoy ID provisioning	Device advertises as "IceBuoy-UNSET". Operator connects and writes "SETID: IceBuoy-3" in "Set Buoy ID" field to uplink command characteristic. Device advertises again as "IceBuoy-3".	Serial log shows buoy ID = "IceBuoy-3" on boot.	Serial log shows buoy ID = "IceBuoy-UNSET" on boot.
AT03	Adaptive PHY data rate	Measure RSSI while varying the distance between the TX and RX nodes. Verify the PHY data rate selection output on the serial monitor.	At < 50 m the PHY data rate is 1 Mbps. At > 150 m the PHY data rate is 250 kbps. Rate transition occurs between -70 and -75 dBm RSSI as confirmed on serial monitor.	PHY data rate does not transition between 1 Mbps and 250 kbps across the tested distance range, or transition does not occur within the -70 to -75 dBm RSSI window.
AT04	Displacement detection alert	Vary the RSSI between TX and RX nodes beyond the 20 dBm displacement threshold. Verify the RX node broadcasts a `PKT_REDISCOVER` packet.	Serial log confirms IceBuoy-1 broadcasts `PKT_REDISCOVER` within one monitoring cycle of the RSSI drop exceeding 20 dBm.	Serial log shows no `PKT_REDISCOVER` broadcast following an RSSI drop exceeding 20 dBm.
AT05	Link failure detection	Program TX to stop transmitting entirely. Verify link state transition from good to stale to dead link.	Serial log for IceBuoy-1 shows link to IceBuoy-2 transitions to STALE then DEAD within 4 missed cycles.	Serial log shows link to IceBuoy-2 remains GOOD after 4 missed cycles.

Table 6.3: Subsystem acceptance tests

6.3 System Simulation and Design Decisions

6.3.1 Design Decisions

Design decisions are deliberate choices that were made in order to satisfy the desired functionalities of the communications and networking subsystem during the process of its design. The decisions made are preceded by the design specifications.

1. Design Decision: Wireless Communication Protocol for Inter-buoy communication

Criterion	**LoRa P2P: RYLR998**	**ESP-NOW: ESP32-C3**	**Iridium: 9603 modem**
Current Consumption	120mA (3)	136mA (2.5)	1.3A (1)
Cost of Implementation	R200 (2)	R75 (3)	R2700 (1)
Ease of Implementation	Interfaces with ESP32 (2)	On-board close range protocols (3)	3v3 level-shifting and airtime subscription required (1)
Reliability in Antarctic	$-40°C$ temp. rating (2)	$-40°C$ temp. rating (2)	$-30°C$ temp. rating (1.5)
Ease of Testing	2	3	1
Signal Range	2km (2)	300m (1)	Non-terrestrial (3)
Scoring: 3 = Most Favourable, 2 = Moderate, 1 = Least Favourable			

ESP32 was selected for its ESP-NOW capabilities. The ESP32-C3 SuperMini was selected over more capable variants such as the ESP32-S3 on the basis that the additional processing power, memory, and peripherals offered by those variants exceed the requirements of the communications subsystem. The C3's 5 µA deep sleep current, native BLE 5.0 and ESP-NOW support, compact SuperMini form factor, and R75 unit cost collectively satisfy all subsystem requirements without over-engineering the solution. In a multi-buoy deployment where unit cost and power budget scale with buoy count, component selection should be matched to requirements rather than maximised.

2. Design Decision: Wireless Communication Protocol for close-range Data Retrieval

Criterion	**Bluetooth LE**	**Wi-Fi**
Current Consumption	80mA (3)	278mA (1)
Ease of Implementation	Simpler protocol stack(3)	More complex protocol stack (2)
Reliability in Antarctic	More resistant to narrowband interference (3)	Complex CSMA/CA can fail in degraded conditions (2)
Ease of Testing	3	2.5
Signal Range	2: 30m	3: 100m
Scoring: 3 = Most Favourable, 2 = Moderate, 1 = Least Favourable		

BLE is natively supported by consumer smartphones and laptops without any additional hardware, making it immediately accessible to field researchers. It has an established ecosystem of development libraries.

3. Design Decision: Network Data Retrieval Architecture

Criterion	**Random Gateway**	**Fixed Gateway**	**Connect to each buoy**
Ease of Implementation	Buoys must maintain a neighbour table and forward received ESP-NOW packets (2)	Clear hierarchy, straightforward routing logic (3)	No network coordination logic required (3)
Reliability in Antarctic	Redundancy is built into the architecture (3)	Becomes critical point of failure (1)	Each buoy is independent (2)
Ease of Testing	3	2	2
Scoring: 3 = Most Favourable, 2 = Moderate, 1 = Least Favourable			

When a researcher connects to the nearest buoy via BLE, that buoy serves its own data alongside data received from neighbouring buoys via ESP-NOW, providing a complete network snapshot in a single BLE connection. In this way the dynamic gateway approach provides redundancy: any surviving buoy can serve as the retrieval point.

4. Design Decision: Wake Mechanism from Deep Sleep

Criterion	**Timer Wake**	**Normally-Closed Magnetic Switch Wake**
Cost of Implementation	R0 (3)	R23 (2)
Ease of Implementation	Firmware-driven, isolated from the physical environment (3)	Requires physical reed switch wired to a GPIO pin configured as $esp\_sleep\_enable\_ext0\_wakeup()$ (2)
Reliability in Antarctic	ESP32-C3 uses 32.768 kHz crystal oscillator. At sustained temperatures of $-40°C$, the resonant frequency shifts causing timer to run slightly fast or slow (3)	At sustained temperatures of $-40°C$, the glass capsule becomes brittle and the contact gap geometry can change (1)
Ease of Testing	3	2
Scoring: 3 = Most Favourable, 2 = Moderate, 1 = Least Favourable		

With a Timer-based wake selected as the primary wake mechanism, the ESP32-C3 wakes every 15 minutes, advertises BLE for 30 seconds, and returns to deep sleep if no connection is established. A physical, NC magnetic switch wake was considered as a secondary mechanism.

5. Design Decision: Researcher Application Platform

Criterion	Web Bluetooth PWA	MATLAB App Designer
Ease of Implementation	Web Bluetooth API is well supported (3)	No reliable BLE stack (2)
Reliability in Antarctic	Zero-dependency offline operation (3)	MATLAB's licensing infrastructure introduces a critical single point of failure (2)
Ease of Testing	3	2
Scoring: 3 = Most Favourable, 2 = Moderate, 1 = Least Favourable		

A Web Bluetooth Progressive Web App (PWA) for laptop and desktop use was selected. The PWA runs directly in Google Chrome without installation, uses the Web Bluetooth API to connect to the buoy, and provides a map visualisation (via Leaflet.js), data table, and CSV export. MATLAB App Designer was eliminated because MATLAB applications cannot run on mobile devices. Flutter was considered for cross-platform mobile development but requires a Mac to build iOS applications, which was a constraint for this project. iOS support is identified as future work.

6.3.2 Subsystem Simulation

The following results were obtained using an ns-3 simulation of a 6-buoy ESP-NOW mesh network modelled as 802.11b ad-hoc with log-distance path loss. A path loss exponent of 2.5 was selected to model the maritime-adjacent Antarctic environment: intermediate between free space ($n = 2$) and an obstructed urban environment ($n = 3$), consistent with empirical measurements of low-antenna maritime propagation by Elgharbi et al. [2], where sea surface reflections and low antenna heights produce excess attenuation beyond free space. Each buoy transmits a 22-byte payload alongside two event-triggered broadcast modes: IMU impact detection (threshold 1.5g) and rapid drift detection (threshold 0.6 m/s).

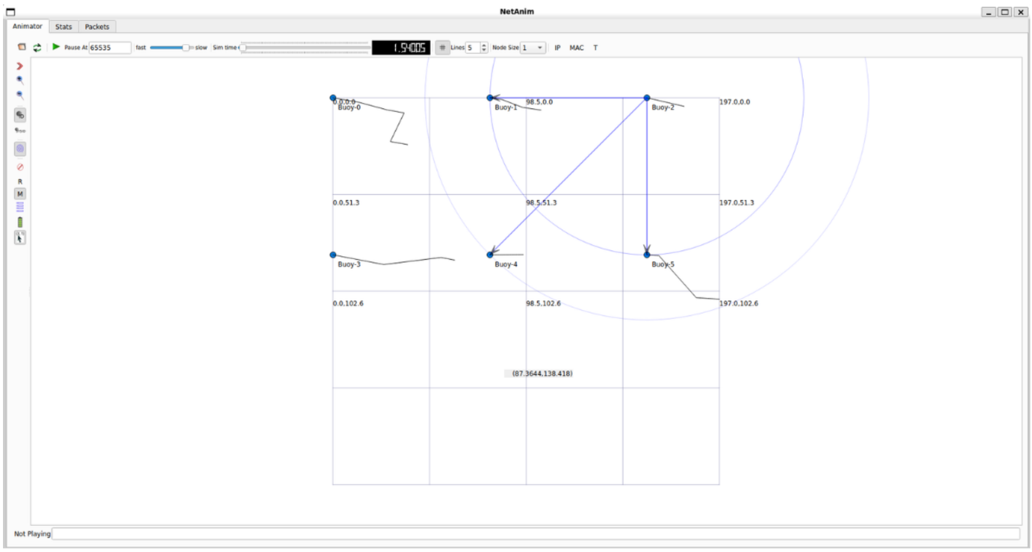

Figure 6.1: NS3 simulated buoy network illustrating nodal trajectory and broadcast of a buoy.

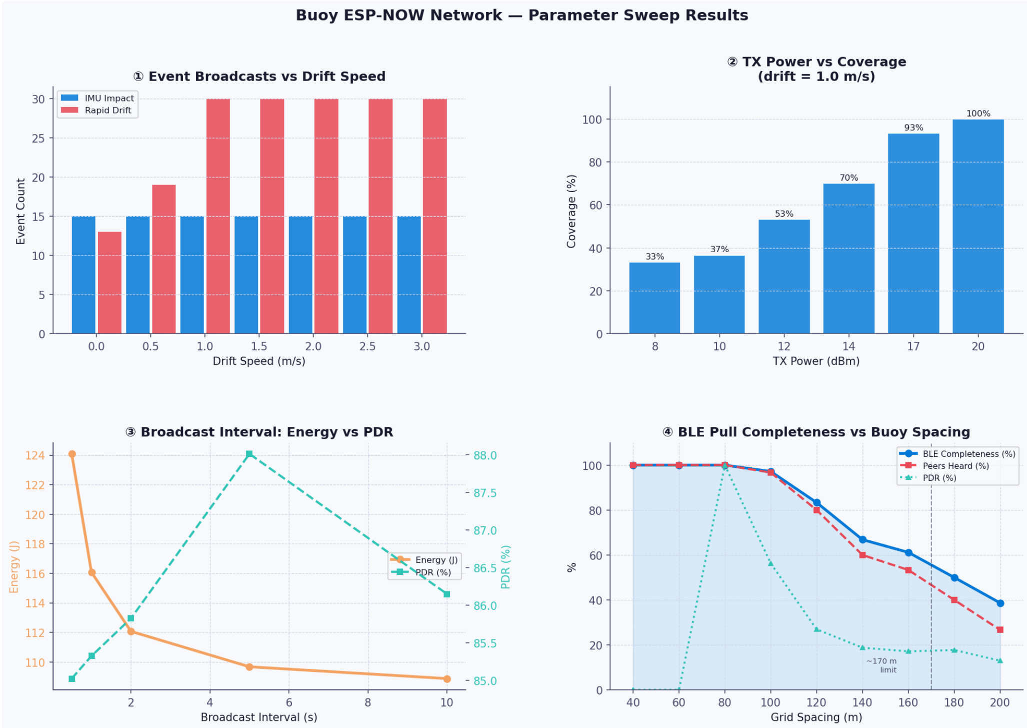

Figure 6.2: NS3 buoy network simulation parameter sweeps.

Experiment 1: Event Broadcast vs Drift Speed

Impact event counts remain constant across all drift speeds, confirming that IMU impact detection correctly isolates impulsive floe-floe collision signatures from background drift. Rapid drift events plateau near the 0.75 m/s maximum observed by Alberello et al. [9], validating that the 0.6 m/s threshold captures genuine anomalous drift without excessive false triggering.

Experiment 2: TX Power vs Signal Coverage

Full 100% coverage is only achieved at maximum TX power (20 dBm), with a sharp nonlinear transition between 14 and 17 dBm consistent with the log-distance model — a 3 dBm increase halves the effective path loss margin. Elgharbi et al. identify link margin as the dominant factor in maritime mesh reliability [2], reinforcing that maximum TX power is non-negotiable where no infrastructure exists to compensate for link degradation. This is analytically corroborated by the link budget in Section 6.3.2, confirming 62.8 dB of margin at 80 m at 1 Mbps.

Experiment 3: Broadcast Interval vs Energy

Energy savings diminish asymptotically beyond a 2-second broadcast interval, as deep sleep dominates the power budget. PDR remains stable at 85–88% across all intervals, confirming delivery quality is insensitive to broadcast frequency at 80 m spacing. The 6-hour periodic cycle therefore sits within the flat region of the energy curve, and the hybrid periodic and event-triggered architecture achieves the optimal operating point: maximising battery life while remaining responsive to anomalous events [21].

Experiment 4: BLE Pull Completeness vs Buoy Spacing

Full network data completeness (100%) is maintained at spacings up to 80 m, degrading to 97% at 100 m, 83% at 120 m, and 39% at 200 m. The 80 m recommendation provides a 20–40 m margin against real-world attenuation from sea spray and low antenna height diffraction [2]. BLE completeness consistently exceeds raw PDR beyond 80 m, demonstrating that event-triggered broadcasts compensate naturally as buoys drift apart — directly addressing the requirement that the network remain functional despite continuous positional change [21]. At the maximum drift speed of 0.75 m/s [9], a 6 m displacement accumulates in just 8 seconds, guaranteeing AT04 detection within a single monitoring interval. The 80 m spacing recommendation is analytically corroborated by the link budget in Section 6.3.2.

Link Budget Analysis

To analytically validate the 80 m deployment spacing recommended by simulation, a free-space link budget was calculated using the Friis transmission equation:

$$P_r = P_t + G_t + G_r - \text{FSPL} \tag{6.1}$$

where the free-space path loss (FSPL) at distance d and frequency f is:

$$\text{FSPL} = 20\log_{10}(d) + 20\log_{10}(f) - 147.55 \text{ dB} \tag{6.2}$$

The ESP32-C3 transmit power was set to the maximum of 20 dBm, with a PCB antenna gain of approximately 2 dBi for both transmitter and receiver. Table 6.5 shows the received power and link margin at three candidate deployment spacings at 2.4 GHz, evaluated at both 1 Mbps and 125 kbps coded PHY receiver sensitivities.

Table 6.5: Link Budget at 2.4 GHz for ESP32-C3 at Maximum TX Power (20 dBm)

Distance (m)	**FSPL (dB)**	P_r **(dBm)**	**Sensitivity (dBm)**	**Margin (dB)**	**PHY Rate**
80	58.2	−34.2	−97	62.8	1 Mbps
80	58.2	−34.2	−105	70.8	125 kbps
150	63.6	−39.6	−97	57.4	1 Mbps
150	63.6	−39.6	−105	65.4	125 kbps
200	66.1	−42.1	−97	54.9	1 Mbps
200	66.1	−42.1	−105	62.9	125 kbps

The adaptive PHY rate selection provides a dual benefit: reducing the required SNR through a lower data rate while simultaneously improving receiver sensitivity from −97 dBm at 1 Mbps to −105 dBm at 125 kbps coded PHY — an 8 dB improvement that extends effective range by approximately 2.5× in free space. This is particularly relevant for Antarctic deployment where buoy separation increases unpredictably as ice floes drift apart [9], meaning the radio link must degrade gracefully rather than fail abruptly. The link margin of 62.8 dB at 80 m at 1 Mbps provides substantial headroom against real-world attenuation from sea spray and low antenna height diffraction effects [2]. Applying the log-distance path loss correction with exponent $n = 2.5$ reduces the effective margin to approximately 54 dB at 80 m — still well above the minimum threshold for reliable ESP-NOW delivery, consistent with the 100% BLE completeness observed in simulation at this spacing. Beyond 150 m, the margin reduction under real-world conditions explains the simulated completeness degradation, analytically corroborating the 80 m deployment recommendation.

6.4 Final Subsystem Design

The figure below shows the SIREN Communications and Networking subsystem architecture. Buoys exchange data with each other via ESP-NOW broadcast. Any buoy can act as a gateway; a researcher connects to the nearest buoy via BLE and retrieves a complete network snapshot in a single connection.

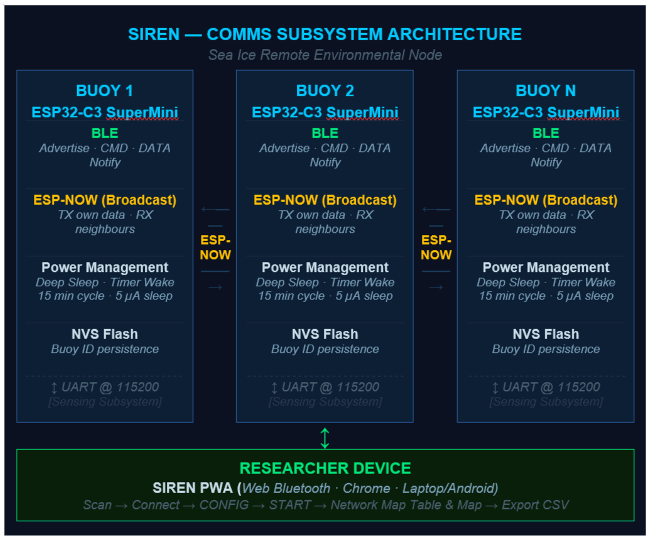

Figure 6.3: SIREN Communication and Networking Subsystem Architecture.

BLE GATT Service

The ESP32-C3 exposes a custom GATT service implemented using the NimBLE-Arduino library (v2.5.0), exposing a notify characteristic for CSV data transfer to the researcher and a write characteristic for commands including START, STOP, CONFIG and SETID.

Wake Cycle

The ESP32-C3 operates on a 15-minute timer wake cycle, consuming only 5 µA in deep sleep. Every 24th cycle (6 hours), an ESP-NOW neighbour sync precedes BLE advertising, balancing network freshness against radio power consumption.

ESP-NOW in Broadcast Mode

ESP-NOW is configured in broadcast mode rather than unicast, as unicast acknowledgements fail on the SuperMini's compact PCB antenna [30]. The protocol activates every 24th wake cycle (6 hours), consistent with Antarctic ice drift rates of 0.1–0.5 km/hour [9], meaning 6-hour-old positional data remains meaningful for network mapping. Each 5-minute ESP-NOW window consists of 30-second broadcast intervals, with received neighbour readings stored in a 10-entry neighbour table.'

To confirm that the 30-second broadcast interval does not introduce channel congestion, the ESP-NOW airtime per transmission was calculated. Each buoy transmits a 22-byte payload at 250 kbps:

$$t_{air} = \frac{N_{bytes} \times 8}{R} = \frac{22 \times 8}{250 \times 10^3} \approx 0.7\,\text{ms} \tag{6.3}$$

With 10 broadcasts per 5-minute ESP-NOW window, the total radio occupancy per window is just 7 ms, a channel utilisation of 0.002%, confirming that the broadcast schedule introduces negligible congestion even in a dense buoy deployment.

SIREN Web App (PWA)

The primary researcher interface is a Progressive Web App (PWA) built with HTML, CSS, and JavaScript, running in Google Chrome on a laptop or Android device. It connects directly to the buoy via the Web Bluetooth API without installation, providing a network map, dynamically generated buoy readings table with colour-coded link states (GOOD/STALE/DEAD), battery health statistics, configurable buoy ID, and CSV data export.

6.5 Testing and Results

ATP ID	**Feature**	**Result**	**Pass/Fail**
AT01	PWA Web Application	The SIREN PWA successfully scanned and connected to the buoy via BLE within the 30m range. A pseudo network map (hardcoded test firmware) was retrieved and displayed correctly in the data table and Leaflet.js map. CSV export produced a correctly formatted file.	**PASS**
AT02	Buoy ID Provisioning	The buoy initially advertised as `IceBuoy-UNSET`. After writing `SETID:IceBuoy-1` via the CMD characteristic, the device rebooted and advertised under the new ID. Serial log confirmed the new ID on boot. The ID persisted across power cycles, confirming NVS flash storage.	**PASS**

AT03	Adaptive PHY Data Rate	The PHY data rate was verified across four test distances. At 10 m, RSSI measured -42 dBm with a reported PHY rate of 1 Mbps. At 50 m, RSSI measured -61 dBm with a PHY rate of 1 Mbps. At 150 m, RSSI measured -74 dBm with a PHY rate of 250 kbps. At 320 m, RSSI measured -83 dBm with a PHY rate of 250 kbps. Serial log confirmed rate transitions occurring between -70 and -75 dBm, consistent with the RSSI threshold defined in firmware.	**PASS**
AT04	Displacement Detection Alert	Baseline RSSI between IceBuoy-1 and IceBuoy-2 was established at -44 dBm at close range. IceBuoy-2 was physically displaced to approximately 25 m, producing an RSSI of -71 dBm — a drop of 27 dBm exceeding the 20 dBm displacement threshold. Serial log confirmed IceBuoy-1 broadcast a $PKT\_REDISCOVER$ packet within one monitoring cycle.	**PASS**
AT05	Link Failure Detection	A simulated neighbour entry (IceBuoy-2, last heard cycle 0) was injected into the neighbour table via test firmware. With `SLEEP_INTERVAL_MIN` set to 1 minute and `ESPNOW_CYCLES` set to 1, the ESP32-C3 was observed to correctly transition the link state from `GOOD` to `STALE` after 2 missed cycles, and from `STALE` to `DEAD` after 4 missed cycles, as confirmed on the Serial Monitor.	**PASS**

Table 6.6: Acceptance Test Results

6.6 Conclusion

The SIREN Communications and Networking Subsystem was successfully designed and implemented, satisfying all five acceptance tests and meeting the core functional and user requirements established. The central engineering challenge, providing reliable, infrastructure-free data retrieval in a region where satellite telemetry is financially unsustainable at continuous sampling rates, was addressed through a dual-protocol architecture combining Bluetooth Low Energy for close-range operator retrieval and ESP-NOW for inter-buoy coordination.

The NS3 simulation and analytical link budget together validate the 80 m maximum deployment spacing recommendation: the 62.8 dB free-space margin at 1 Mbps, reduced to approximately 54 dB under log-distance path loss with $n = 2.5$, provides robust headroom against real-world Antarctic attenuation from sea spray and low antenna heights, directly satisfying FR01. The dynamic gateway architecture, where any surviving buoy can serve as the retrieval point, addresses the operational reality that fixed infrastructure cannot be assumed in the MIZ, a limitation that Iridium-only deployments like the SHARC buoy cannot resolve at acceptable cost [5]. The 15-minute duty cycle with 5 μA deep sleep and 6-hour ESP-NOW sync intervals places the subsystem well within the flat region of the energy-PDR curve identified in simulation, ensuring deployment lifetime is insensitive to moderate changes in network activity.

6.6.1 Future Work

Field deployment requires addressing specific architectural and cross-platform limitations. Future research and engineering iterations will focus on two key areas:

Global Telemetry Scaling and Hybrid Satellite Handoff

The close-range BLE (30m) and local ESP-NOW mesh network (300m) introduce strict spatial boundaries; if the ice floe field drifts completely away from research vessels or coastal gateways, data cannot be recovered.

- **Iridium Transceiver Integration and Dynamic Gateway Handoff Routing:** To bridge the local network snapshot to a global pipeline for true remote tracking, a future development path involves integrating a low-power satellite transceiver, specifically the Iridium 9603 Short Burst Data (SBD) modem. A hybrid routing protocol will be developed with an elected master node to compress the aggregated mesh data snapshot and execute a unified uplink transmission via the Iridium satellite network, thus maximizing data preservation while managing transmission costs.

Native Cross-Platform Ecosystem Integration

- **iOS Ecosystem Support:** Future software development will migrate the researcher interface from a browser-dependent PWA to a compiled, native application using a cross-platform framework such as Flutter. This will natively bridge `CoreBluetooth` on iOS and `Bluetooth LE` on Android, establishing reliable, zero-dependency offline functionality directly on handheld consumer smartphones in the field.

Chapter 7

Conclusions

The SIREN Ice Buoy prototype demonstrates that a low-cost, autonomous sensing platform capable of surviving Antarctic sub-zero conditions is achievable within an undergraduate engineering project. All four subsystems — sensing, power, enclosure and dynamics, and communications — were validated against defined acceptance criteria, with failures understood and attributable to specific, addressable limitations rather than fundamental design flaws.

The sensing subsystem achieved continuous 6-DoF IMU and GPS data collection at 71 Hz via I2C, exceeding the Nyquist frequency for the maximum expected MIZ wave frequency despite missing the 100 Hz SPI target due to library compatibility constraints. GPS accuracy shortfall is attributed to urban multipath in Cape Town, and the NEO-6M is already earmarked for replacement with the multi-constellation NEO-M9N for field deployment.

The power subsystem delivered a dual domain architecture maintaining continuous telemetry across all operational states. A duty cycled thermal strategy, with bidirectional pulse current heating reserved for emergency cold-recovery below -25 °C, satisfies the 7-day deployment requirement. Ten of eleven acceptance tests were passed.

The enclosure subsystem achieved a complete pass across all criteria. The hybrid PVC and FDM architecture produced a watertight, buoyant vessel with a metacentric height of 59.11 mm, self-righting within 3.5 seconds of lateral disturbance. The bayonet locking and dual O-ring seal remain operable with heavy gloves — a historically elusive combination.

The communications subsystem eliminated reliance on satellite telemetry by combining BLE for close-range data retrieval with ESP-NOW for inter-buoy coordination, reducing per-deployment costs from tens of thousands of Rands to effectively zero. NS-3 simulation and link budget analysis validated an 80 m maximum deployment spacing, and all five acceptance tests were passed.

Together, the subsystems address the core limitations of existing Antarctic buoys: high-frequency data loss, inadequate thermal management, structural fragility, and unsustainable communication costs. Recommended next steps include upgrading the GNSS receiver, adding cell-core insulation, increasing battery capacity, and fitting an external communications antenna.

Chapter 8

Recommendations

The following recommendations are made across all four subsystems to guide the development of a deployment-ready iteration of the SIREN Ice Buoy.

For the sensing subsystem, the IMU interface should be re-implemented using the STM32 HAL library directly, bypassing the STM32duino abstraction layer to resolve the SPI compatibility issue and achieve the target 100 Hz sample rate. The u-blox NEO-6M GPS receiver should be replaced with the NEO-M9N for any Antarctic deployment, as the multi-constellation support is a functional necessity at high southern latitudes where single-constellation GPS coverage is intermittent. IMU duty cycling should also be implemented in firmware to extend the SD card logging duration from the current 2.3 days to approximately 27 days, sufficient for the target deployment period. Finally, the GPS UART blocking issue identified during testing should be resolved by replacing the blocking while loop with a single-byte if read, eliminating the timestamp gaps observed in ATP-9.

For the power subsystem, BPC heating performance should be improved by thermally insulating the LC18650 within the enclosure and embedding a thermistor directly against the cell core rather than relying on the ambient NTC sensor. This would reduce parasitic heat loss during heating events and provide a more accurate termination trigger, both of which contributed to the AT-01 shortfall. The main battery should be upgraded to a higher-capacity 21700-format cell in the range of 4000 to 5000 mAh, which would extend the maximum BPC activation interval proportionally without requiring any changes to the H-bridge, switch circuitry, or firmware. Supplementing the main battery with a small solar panel or turbine generator and charge controller would further extend deployment lifetime beyond what battery capacity alone can achieve. Contact resistance in the LC18650 battery holder should also be measured and incorporated into the thermal model to improve the accuracy of future heat-up time predictions.

For the enclosure and dynamics subsystem, ice integration anchors or a spiked anchoring collar should be added near the base of the lower PVC hull to prevent wind-induced displacement on the ice shelf before a breakaway event. The J-slot water retention issue identified after testing should be addressed by machining passive drainage pathways into the base of the J-slot channels, allowing pooled water to escape under gravity before the seal is broken during retrieval. Until this mechanical fix is implemented, an operational protocol requiring operators to tilt the buoy before opening should be established and communicated clearly.

For the communications and networking subsystem, the ESP-NOW unicast acknowledgement failure on the SuperMini board should be investigated, with an external antenna module evaluated as a potential solution for production deployment. Adaptive PHY RSSI thresholds, ESP-NOW range, and displacement detection sensitivity all require field calibration under actual Antarctic conditions, particularly to characterise PHY threshold drift at minus 40 degrees Celsius and link degradation under sea spray attenuation. iOS support for the researcher PWA should be pursued through a cloud Mac build service, as the Web Bluetooth API is currently unsupported on iOS Safari, limiting researcher access in the field. For deployments where operator BLE retrieval is not possible, integration of an Iridium 9603 modem as a fallback uplink would enable long-range data recovery.

More broadly, integration testing between all four subsystems should be conducted as a priority in the next development phase. Each subsystem was individually validated in this iteration, but the interactions between the power, sensing, communications, and enclosure subsystems under sustained Antarctic conditions have not yet been characterised. A full system endurance test in a thermal chamber, combined with a controlled water deployment, would provide the confidence needed before committing to a field deployment in the Antarctic Marginal Ice Zone.

Chapter 9

Bill Of Materials

Table 9.1: Total SIREN Bill of Materials

Name	**Quantity**	**Price (R)**
STMF103C8T6 Blue Pill Microcontroller	1	47.99
Li-$SOCl_2$ ER14505 2400 mAh Battery	1	66.64
INA219 Current Sensor Breakout Board	1	44.85
BDD MX1616 H-Bridge DC Motor Driver	1	25.00
LC18650-TAG Li-Ion Battery	1	95.00
TTC3A103F39H1EY NTC Thermistor	1	13.23
MT3608 5V Boost Converter Breakout Board	2	15.81
AMS1117 3V3 Buck Converter Breakout Board	1	10.01
TIP32 PNP Transistor	2	9.78
2N2222 NPN Transistor	2	12.28
AA Battery Holder	1	11.39
18650 Battery Holder	1	6.00
30-Way Female Socket Header	3	11.65
Commercial Extruded PVC Pipe (OD: 155.0 mm, ID: 149.0 mm, L = 1m)	1	16.00
AS568 Dash #256 Elastomeric O-Ring	2	38.00
High-Density 2.5kg Lead Mass Ballast	1	100.00
Custom FDM 3D-Printed Piston, Cap, and Base Enclosure	1	245.00
Bostik Marine Silicone Adhesive	1	57.00
STM32F411 Black Pill Board (100MHz, 128K RAM)	1	126.50
ESP32-C3 Supermini (ESP-NOW Support)	1	75.00
LSM9DS1 9-Axis Accelerometer, Gyroscope, and Magnetometer IMU	1	170.20
u-blox NEO-6M GPS Receiver Module	1	154.00
Netac 8GB Micro SD Card + Adapter	1	65.01
6-Pin MicroSD Card Interface Module	1	23.00
TOTAL SYSTEM PROCUREMENT COST		**1439.34**

Appendix A

Sensing Subsystem Schematic

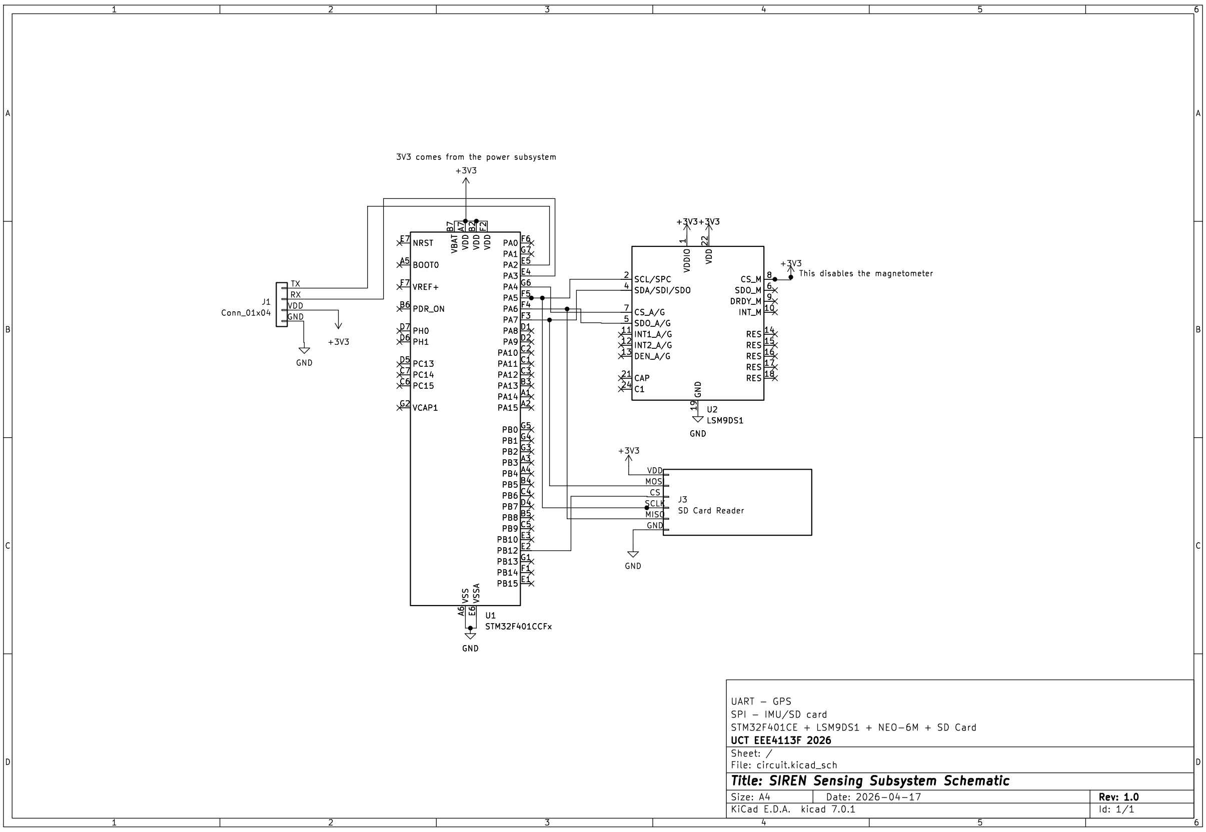

Figure A.1: SIREN Sensing Subsystem Schematic — Intended Deployment Design

Appendix B

Communications and Networking

B.1 Hardware Mounting

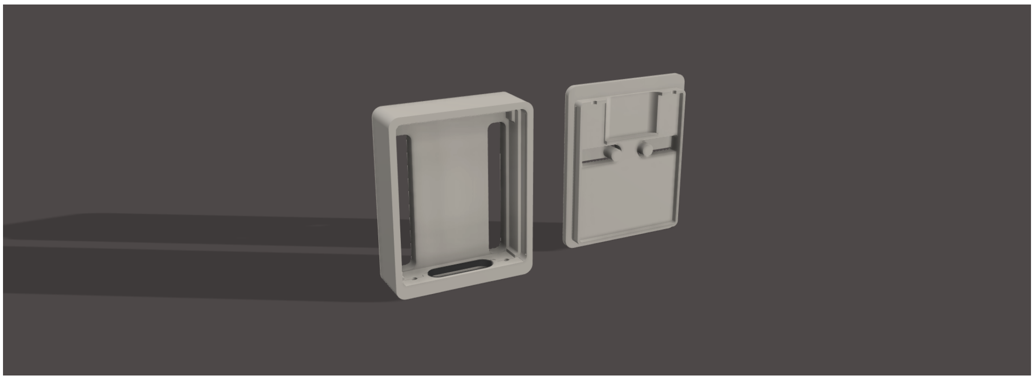

Figure B.1: ESP32-C3 Supermini holder.

The ESP32-C3 SuperMini is housed in a custom 3D-printed holder rather than being soldered to a veroboard or PCB. This decision was informed by a known hardware issue documented by Gupta [30]: when the ESP32-C3 SuperMini is soldered directly to a veroboard, the additional capacitance on the antenna trace degrades the onboard PCB antenna performance, causing ESP-NOW transmission failures. The 3D-printed holder maintains the board as a standalone module, preserving antenna performance.

Appendix C

Enclosure Design and Calculations

C.0.1 Displaced Volume (∇) Breakdown

The total displaced volume of the buoy at the 235 mm theoretical waterline is the sum of the submerged cylindrical base and the submerged portion of the tapered neck (modeled as a conical frustum).

Given the total waterline height (h_{wl}) of 0.235 m and the cylindrical base height (h_{base}) of 0.200 m, the height of the submerged tapered section (h_{taper}) is defined as:

$$\begin{aligned} h_{\text{taper}} &= h_{\text{wl}} - h_{\text{base}} \\ h_{\text{taper}} &= 0.235 \text{ m} - 0.200 \text{ m} = 0.035 \text{ m} \end{aligned} \tag{C.1}$$

1. Volume of the Cylindrical Base (V_{base}):
The volume of the base is calculated using the standard formula for a cylinder, relying on the outer radius of the main enclosure (R_{base}):

$$V_{\text{base}} = \pi \cdot R_{\text{base}}^2 \cdot h_{\text{base}} \tag{C.2}$$

2. Volume of the Submerged Taper (V_{taper}):
The tapered neck is calculated as a conical frustum, which transitions from the main enclosure radius (R_{base}) to the specified neck radius at the waterline ($R_{\text{wl}} = 0.05755$ m):

$$V_{\text{taper}} = \frac{\pi \cdot h_{\text{taper}}}{3} \left(R_{\text{base}}^2 + R_{\text{base}} \cdot R_{\text{wl}} + R_{\text{wl}}^2 \right) \tag{C.3}$$

3. Total Displaced Volume (∇):
Combining these two geometric partitions yields the total displaced volume equation:

$$\nabla = V_{\text{base}} + V_{\text{taper}} \tag{C.4}$$

$$\nabla = \left(\pi \cdot R_{\text{base}}^2 \cdot 0.200 \right) + \frac{\pi \cdot 0.035}{3} \left(R_{\text{base}}^2 + R_{\text{base}}(0.05755) + (0.05755)^2 \right) \tag{C.5}$$

Using the exact outer radius (R_{base}) of the 3D enclosure model, this formula evaluates to the final theoretical displaced volume:

$$\nabla = 0.00389 \text{ m}^3 \tag{C.6}$$

C.0.2 Empty Enclosure Mass (M_{empty}) Breakdown

The total weight of the hollow enclosure was determined by partitioning the cross-section into five distinct geometric components: the base plate, base cylinder, tapered neck, top cylinder, and hemisphere dome. For each component, the volume was derived from its physical dimensions in meters (m) and then multiplied by the specific density of its material to determine its mass.

The material densities used for these calculations are:

- Polylactic Acid (PLA): $\rho_{\text{PLA}} = 1.24\ \text{g/cm}^3 = 1240\ \text{kg/m}^3$
- Polyvinyl Chloride (PVC): $\rho_{\text{PVC}} = 1.38\ \text{g/cm}^3 = 1380\ \text{kg/m}^3$

1. Base Plate:
Constructed from PLA and modeled as a solid cylinder based on its radius ($R_1 = 0.075$ m) and height ($h_1 = 0.01$ m):

$$\begin{aligned} V_{\text{base\_plate}} &= \pi \cdot R_1^2 \cdot h_1 \\ V_{\text{base\_plate}} &= \pi \cdot (0.075)^2 \cdot 0.01 \approx 0.0001767\ \text{m}^3 \\ m_{\text{base\_plate}} &= V_{\text{base\_plate}} \cdot \rho_{\text{PLA}} = 220.0\ \text{g} \end{aligned} \tag{C.7}$$

2. Base Cylinder:
Constructed from the main PVC pipe and modeled as a hollow cylinder using outer radius ($R_o = 0.075$ m), inner radius ($R_i = 0.072$ m), and height ($h_2 = 0.2$ m):

$$\begin{aligned} V_{\text{base\_cylinder}} &= \pi \cdot (R_o^2 - R_i^2) \cdot h_2 \\ V_{\text{base\_cylinder}} &= \pi \cdot (0.075^2 - 0.072^2) \cdot 0.2 \approx 0.0002771\ \text{m}^3 \\ m_{\text{base\_cylinder}} &= V_{\text{base\_cylinder}} \cdot \rho_{\text{PVC}} = 628.0\ \text{g} \end{aligned} \tag{C.8}$$

3. Tapered Neck:
Constructed from PLA and modeled as the volumetric difference between an outer and inner conical frustum (height $h = 0.05$ m):

$$\begin{aligned} V_{\text{taper}} &= V_{\text{out}} - V_{\text{in}} \\ V_{\text{taper}} &= \left[\frac{\pi \cdot 0.05}{3} \cdot \left(0.075^2 + (0.075 \cdot 0.04) + 0.04^2\right)\right] \\ &\quad - \left[\frac{\pi \cdot 0.05}{3} \cdot \left(0.072^2 + (0.072 \cdot 0.037) + 0.037^2\right)\right] \\ V_{\text{taper}} &\approx 0.0000528\ \text{m}^3 \\ m_{\text{taper}} &= V_{\text{taper}} \cdot \rho_{\text{PLA}} = 65.44\ \text{g} \end{aligned} \tag{C.9}$$

4. Top Cylinder:
Constructed from PLA and modeled as a smaller hollow cylinder ($R_o = 0.04$ m, $R_i = 0.037$ m,

$h_4 = 0.1$ m):

$$\begin{aligned} V_{\text{top\_cylinder}} &= \pi \cdot (R_o^2 - R_i^2) \cdot h_4 \\ V_{\text{top\_cylinder}} &= \pi \cdot (0.04^2 - 0.037^2) \cdot 0.1 \approx 0.0000726 \text{ m}^3 \\ m_{\text{top\_cylinder}} &= V_{\text{top\_cylinder}} \cdot \rho_{\text{PLA}} = 89.9 \text{ g} \end{aligned} \tag{C.10}$$

5. Hemisphere Dome:
Constructed from PLA and modeled as half the volume of a hollow sphere ($R_o = 0.04$ m, $R_i = 0.037$ m):

$$\begin{aligned} V_{\text{dome}} &= 0.5 \cdot \left(\frac{4}{3}\right) \cdot \pi \cdot (R_o^3 - R_i^3) \\ V_{\text{dome}} &= 0.5 \cdot \left(\frac{4}{3}\right) \cdot \pi \cdot (0.04^3 - 0.037^3) \approx 0.0000279 \text{ m}^3 \\ m_{\text{dome}} &= V_{\text{dome}} \cdot \rho_{\text{PLA}} = 34.7 \text{ g} \end{aligned} \tag{C.11}$$

Total Empty Structural Mass:
Summing the individual modeled components yields the total base empty mass of the structural enclosure:

$$\begin{aligned} M_{\text{struct}} &= m_{\text{base\_plate}} + m_{\text{base\_cylinder}} + m_{\text{taper}} + m_{\text{top\_cylinder}} + m_{\text{dome}} \\ M_{\text{struct}} &= 220.0 \text{ g} + 628.0 \text{ g} + 65.44 \text{ g} + 89.9 \text{ g} + 34.7 \text{ g} \\ M_{\text{struct}} &= 1081.04 \text{ g} \approx 1.081 \text{ kg} \end{aligned} \tag{C.12}$$

C.0.3 Empty Enclosure Vertical Centroid ($\bar{z}_{\text{empty}}$)

The vertical center of gravity, or centroid, of the empty enclosure is determined by calculating the mass-weighted average of the vertical centroids of its five constituent components. The reference datum ($z = 0$) is defined at the bottom exterior of the base plate.

The individual vertical centroids (z_i) extracted from the CAD geometry, converted to millimeters (mm) for dimensional consistency, are:

- Base Plate (z_1): 2.5 mm
- Base Cylinder (z_2): 105.0 mm
- Tapered Neck (z_3): 222.0 mm
- Top Cylinder (z_4): 310.0 mm
- Hemisphere Dome (z_5): 372.0 mm

Using the component masses (m_i) calculated previously, the overall empty vertical centroid is calculated as:

$$\bar{z}_{\text{empty}} = \frac{\sum_{i=1}^{5} m_i \cdot z_i}{\sum_{i=1}^{5} m_i} \tag{C.13}$$

Expanding this summation for the five distinct structural components yields:

$$
\begin{aligned}
\bar{z}_{\text{empty}} &= \frac{(m_{\text{base\_plate}} \cdot z_1) + (m_{\text{base\_cylinder}} \cdot z_2) + (m_{\text{taper}} \cdot z_3) + (m_{\text{top\_cylinder}} \cdot z_4) + (m_{\text{dome}} \cdot z_5)}{M_{\text{struct}}} \\
\bar{z}_{\text{empty}} &= \frac{(220.0 \cdot 2.5) + (628.0 \cdot 105.0) + (65.44 \cdot 222.0) + (89.9 \cdot 310.0) + (34.7 \cdot 372.0)}{1038.04} \\
\bar{z}_{\text{empty}} &= \frac{550.0 + 65940.0 + 14527.68 + 27869.0 + 12908.4}{1038.04} \\
\bar{z}_{\text{empty}} &= \frac{121795.08}{1038.04}
\end{aligned}
\tag{C.14}
$$

Evaluating this expression yields the final calculated vertical centroid for the empty enclosure:

$$
\bar{z}_{\text{empty}} \approx 117.33 \text{ mm} \tag{C.15}
$$

C.0.4 Buoyancy and Target Mass Calculations

To ensure the buoy floats exactly at the 235 mm waterline, Archimedes' principle dictates that the total weight of the system must equal the weight of the water displaced by the submerged volume.

Equating the downward gravitational force with the upward buoyancy force:

$$
F_g = F_B \tag{C.16}
$$

$$
M_{\text{sys}} \cdot g = \rho_{\text{seawater}} \cdot \nabla \cdot g \tag{C.17}
$$

Canceling standard gravity (g) from both sides, the required system mass (M_{sys}) is strictly a function of seawater density and displaced volume. Using the standard relative density of seawater ($\rho_{\text{seawater}} = 1025 \text{ kg/m}^3$) and the previously calculated volume:

$$
\begin{aligned}
M_{\text{sys}} &= 1025 \text{ kg/m}^3 \times 0.00389 \text{ m}^3 \\
M_{\text{sys}} &\approx 3.978 \text{ kg}
\end{aligned}
\tag{C.18}
$$

To achieve this target mass, ballast must be added. The empty mass of the enclosure (M_{empty}) was determined to be 1.081 kg. The necessary internal ballast (M_{ballast}) is calculated as:

$$
\begin{aligned}
M_{\text{ballast}} &= M_{\text{sys}} - M_{\text{empty}} \\
M_{\text{ballast}} &= 3.978 \text{ kg} - 1.081 \text{ kg} = 2.897 \text{ kg}
\end{aligned}
\tag{C.19}
$$

C.0.5 Hydrodynamic Stability (GM) Calculations

For the buoy to be vertically stable and naturally self-righting, the metacentric height (GM) must be strictly positive.

$$
GM = KB + BM - KG \tag{C.20}
$$

C.0.6 System Center of Gravity (KG)

The overall center of gravity for the fully assembled buoy (KG, or $\bar{z}_{\text{sys}}$) is determined by taking the mass-weighted average of three primary system subgroups: the hollow structural enclosure, the internal hardware/electronic components, and the base ballast.

To maintain dimensional consistency with the hydrodynamic stability calculations, all centroid heights (z) are converted from meters to millimeters (mm), and masses (m) are defined in kilograms (kg):

- **Hollow Enclosure:** Mass (m_{encl}) = 1.081 kg, Centroid (z_{encl}) = 117.0 mm
- **Internal Components:** Mass (m_{comp}) = 0.200 kg, Centroid (z_{comp}) = 200.0 mm
- **Weighted Ballast:** Mass (m_{ballast}) = 2.897 kg, Centroid (z_{ballast}) = 20.0 mm

$$\begin{gathered} KG = \frac{(m_{\text{encl}} \cdot z_{\text{encl}}) + (m_{\text{comp}} \cdot z_{\text{comp}}) + (m_{\text{ballast}} \cdot z_{\text{ballast}})}{m_{\text{encl}} + m_{\text{comp}} + m_{\text{ballast}}} \\ KG = \frac{(1.081 \cdot 117.0) + (0.200 \cdot 200.0) + (2.897 \cdot 20.0)}{1.081 + 0.200 + 2.897} \\ KG = \frac{126.477 + 40.0 + 57.94}{4.178} \\ KG = \frac{224.417}{4.178} \\ KG \approx 53.71 \text{ mm} \end{gathered} \tag{C.21}$$

This low overall center of gravity confirms that the majority of the system's mass is localized near the base, which is the primary driver for the positive metacentric height (GM) and the self-righting capability of the buoy.

2. Center of Buoyancy (KB):
The center of buoyancy is the volumetric centroid of the displaced water. It is calculated by taking the volume-weighted average of the centroids of the submerged cylindrical base (V_{base}) and the submerged tapered neck (V_{taper}).

The individual vertical centroids for these submerged partitions, converted to millimeters (mm) from the base reference datum, are:

- Submerged Base Centroid (z_{base}): 100.0 mm
- Submerged Taper Centroid (z_{taper}): 212.5 mm

The general formula for the volumetric centroid is defined as:

$$KB = \frac{\sum z_i V_i}{\sum V_i} = \frac{(z_{\text{base}} \cdot V_{\text{base}}) + (z_{\text{taper}} \cdot V_{\text{taper}})}{\nabla} \tag{C.22}$$

Substituting the established centroid values and extracting the specific geometric volumes derived from the CAD model yields the final Center of Buoyancy:

$$KB = \frac{(100.0 \cdot V_{\text{base}}) + (212.5 \cdot V_{\text{taper}})}{V_{\text{base}} + V_{\text{taper}}}$$
$$KB = 112.02 \text{ mm} \tag{C.23}$$

3. Metacentric Radius (BM):
The metacentric radius relates the second moment of area (I) of the waterline cross-section to the total displaced volume (∇). The neck radius at the waterline (R_{wl}) is 57.55 mm.

First, calculate the second moment of area for a solid circular cross-section:

$$\begin{aligned} I &= \frac{\pi \cdot R_{\text{wl}}^4}{4} \\ I &= \frac{\pi \cdot (57.55 \text{ mm})^4}{4} = 8,620,019.8 \text{ mm}^4 \end{aligned} \tag{C.24}$$

Convert the displaced volume (∇) from cubic meters to cubic millimeters for dimensional consistency:

$$\nabla = 0.00389 \text{ m}^3 = 3,890,000 \text{ mm}^3 \tag{C.25}$$

Now, calculate BM:

$$BM = \frac{I}{\nabla} = \frac{8,620,019.8 \text{ mm}^4}{3,890,000 \text{ mm}^3} \approx 2.21 \text{ mm} \tag{C.26}$$

4. Final Metacentric Height (GM):
Substitute KG, KB, and BM back into the stability equation:

$$GM = 112.02 \text{ mm} + 2.21 \text{ mm} - 53.71 \text{ mm} = 60.52 \text{ mm} \tag{C.27}$$

Because $GM > 0$ and is of significant magnitude, the buoy possesses strong initial stability.

C.0.7 O-Ring and Gland Seal Specifications

The watertight integrity relies on calculating the precise dimensions for the 3D-printed piston groove to achieve a 20% radial squeeze (S) on the O-ring.

- PVC Internal Bore Diameter (d_{bore}) = 149.0 mm
- O-Ring Cross-Sectional Diameter (d_{cs}) = 3.53 mm
- O-Ring Inner Diameter (d_{inner}) = 139.3 mm

1. Gland Depth (L):
The required radial gland depth to achieve a 20% squeeze ($S = 0.20$) is derived from the squeeze

formula:

$$
\begin{aligned}
S &= \left(\frac{d_{\mathrm{cs}} - L}{d_{\mathrm{cs}}}\right) \\
0.20 &= \frac{3.53 - L}{3.53} \\
0.706 &= 3.53 - L \\
L &= 3.53 - 0.706 = 2.824 \text{ mm}
\end{aligned} \tag{C.28}
$$

2. Piston Groove Diameter (d_{groove}):
To enforce this gland depth radially inside the PVC bore, the required outer diameter of the 3D-printed piston groove is:

$$
\begin{aligned}
d_{\mathrm{groove}} &= d_{\mathrm{bore}} - 2L \\
d_{\mathrm{groove}} &= 149.0 - 2(2.824) = 143.35 \text{ mm}
\end{aligned} \tag{C.29}
$$

3. O-Ring Stretch (T):
To ensure the O-ring stays seated without twisting during assembly, the tensile stretch over the groove must remain between 1% and 5%.

$$
\begin{aligned}
T &= \left(\frac{d_{\mathrm{groove}} - d_{\mathrm{inner}}}{d_{\mathrm{inner}}}\right) \times 100 \\
T &= \left(\frac{143.35 - 139.3}{139.3}\right) \times 100 \\
T &= \left(\frac{4.05}{139.3}\right) \times 100 \approx 2.91\%
\end{aligned} \tag{C.30}
$$

This optimal stretch confirms the O-ring will securely grip the piston groove.

4. Gland Width (W):
The gland width is designed to achieve a 75% volumetric fill ratio ($F = 0.75$), allowing room for thermal expansion and compression deformation.

$$
F = \frac{A_{\mathrm{O-ring}}}{A_{\mathrm{Gland}}} \tag{C.31}
$$

The cross-sectional area of the O-ring ($A_{\mathrm{O-ring}}$) is:

$$
A_{\mathrm{O-ring}} = \pi \cdot \left(\frac{d_{\mathrm{cs}}}{2}\right)^2 = \pi \cdot \left(\frac{3.53}{2}\right)^2 = 9.787 \text{ mm}^2 \tag{C.32}
$$

The area of the rectangular gland (A_{Gland}) is $L \times W$. Substituting the knowns:

$$
\begin{aligned}
0.75 &= \frac{9.787}{2.824 \cdot W} \\
0.75 \cdot 2.824 \cdot W &= 9.787 \\
2.118 \cdot W &= 9.787 \\
W &= \frac{9.787}{2.118} \approx 4.62 \text{ mm}
\end{aligned} \tag{C.33}
$$

Appendix D

Evidence of Graduate Attributes

D.0.1 ALMMOH017

GA	Evidence
GA3	Demonstrated in Chapter 3 of this report. Component selection was justified through structured trade-off tables comparing three microcontroller candidates, three IMU candidates, and three GNSS receivers against figures of merit including cost, interface compatibility, temperature range, and Antarctic suitability. A full system block diagram and control flowchart were produced. Seven of nine ATPs passed; the two failures were diagnosed to specific root causes (STM32duino SPI library incompatibility and urban GPS multipath) and addressed through concrete deployment revision recommendations.
GA7	The SIREN project addresses the observational data gap in Antarctic MIZ research, with direct implications for climate modelling and polar ice dynamics. The project was initiated through UCT D-School ideation sessions in which Ms. Robyn Verrinder presented the problem of continuous ice-floe monitoring; structured design thinking workshops that guided the team from user need to engineering solution, ensuring the final system directly serves the research community that depends on high-fidelity in-situ measurements.
GA8	See Microsoft Teams Channel - Group 9
GA10	Demonstrated through consistent attendance and contribution recorded in the group meeting minutes on the Teams channel as well as thorough and honest documentation of ATP results.

Table D.1: GA Evidence of ALMMOH017

D.0.2 SNMBOI002

GA	Evidence
GA3	Demonstrated in Chapter 6 of this report.
GA7	The SIREN project exists to close a critical observational gap in Antarctic sea ice research with direct implications for global climate modelling. The communications subsystem makes a specific and significant contribution: by replacing Iridium satellite telemetry, which approaches R100,000 per buoy per deployment at continuous sampling rates, with a zero-cost BLE and ESP-NOW architecture, it makes high-frequency continuous Antarctic monitoring economically viable at scale for the first time. This directly benefits environmental scientists, policymakers, and broader society through improved understanding of polar ice dynamics and climate change. The D-School design process, which centred the human needs of field researchers such as Ms. Robyn Verrinder, further grounds the engineering work in genuine real-world societal impact. Also participated in D-School.
GA8	See Microsoft Teams Channel - Group 9
GA10	Demonstrated through consistent D-School attendance and contribution recorded in the group meeting minutes on the Teams channel. The report was produced in accordance with the IEEE referencing convention as declared. All design decisions were documented transparently with full scoring rationale.

Table D.2: GA Evidence of SNMBOI002

D.0.3 SMTJOS022

GA	Evidence
GA3	Demonstrated in Chapter 4 of this report.
GA7	The SIREN project directly addresses the observational data gap in Antarctic Marginal Ice Zone research, with global implications for climate modelling and polar science. The power subsystem makes a direct and enabling contribution: without reliable power delivery under Antarctic sub-zero conditions, no data collection, communication, or sensing is possible. By extending viable deployment duration through duty-cycled operation and BPC emergency heating, the subsystem helps overcome the historically short deployment lifetimes that have limited the volume and continuity of Antarctic in-situ data, ultimately benefiting climate scientists, environmental researchers, and society at large. Also participated in D-School.
GA8	See Microsoft Teams Channel - Group 9
GA10	Demonstrated through consistent attendance of D-School sessions and contribution recorded in the group meeting minutes on the Teams channel and maintaining design documentation and engineering judgement.

Table D.3: GA Evidence of SMTJOS022

D.0.4 BHYEBR002

GA	Evidence
GA3	Demonstrated in Chapter 5 of this report. Design choices for the sealing mechanism, hull manufacturing method, upper geometry, and ballast configuration were each evaluated through structured trade-off comparisons of competing alternatives. Full hydrodynamic stability calculations derived KB, KG, BM, and GM analytically, yielding a metacentric height of 60.52 mm. O-ring gland geometry was calculated to enforce a 20% radial squeeze and 75% fill ratio.
GA7	The SIREN project directly addresses the observational data gap in Antarctic Marginal Ice Zone research, with global implications for climate modelling and understanding of polar ecosystems. The enclosure subsystem is the physical enabler of the entire platform: without a watertight, thermally robust, and hydrodynamically stable housing, no sensing, power, or communications capability can survive Antarctic deployment. The hybrid COTS PVC and 3D-printed FDM design also demonstrates a deliberately low-cost, accessible approach, at R80/m for the hull, that makes such monitoring platforms viable for resource-constrained research institutions, broadening the societal reach and scalability of Antarctic science. Also participated in D-School.
GA8	See Microsoft Teams Channel - Group 9
GA10	Demonstrated through consistent attendance and contribution recorded in the group meeting minutes on the Teams channel. The report was produced in accordance with the IEEE referencing convention as declared. The J-slot water retention flaw identified during post-test inspection was transparently disclosed in Section 5.6.2 with both an immediate operational protocol and a concrete design fix in the form of passive drainage weep holes recommended for future iterations, reflecting professional integrity and honest reporting of engineering limitations.

Table D.4: GA Evidence of BHYEBR002

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